Bigtech Credit Market Competition and Bank Portfolio Risk*

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Abstract

What happens to incumbent banks when BigTech platforms use information gathered outside finance to compete for their borrowers? To explore this, I develop a sequential credit-market model in which a bank chooses costly screening and a uniform loan rate before competing with a digital platform whose information improves with ecosystem adoption. I show that platform entry can increase the bank's portfolio risk by disproportionately attracting safe borrowers, even without increasing risky lending. In equilibrium, the bank raises its loan rate while adjusting screening only moderately, as defending contestable borrowers requires repricing existing loans. Improvements in platform information strengthen this adverse selection, while funding costs shape the extent of competition. Despite deteriorating bank portfolio quality, platform entry can expand productive lending and improve aggregate efficiency. These findings highlight how information-driven competition can have contrasting effects on incumbent banks and aggregate credit allocation.

Keywords: Banking, Credit Market Competition, Digital Platforms

JEL Codes: G21, G23, L1

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1 Introduction

The growing dominance of digital platforms and Bigtech firms across various sectors has raised significant concerns regarding their impact on competition. These firms leverage their vast stocks of user data, cutting-edge technologies, and network effects to expand into markets historically served by regulated incumbents, with one of them being, the financial sector.

Unlike conventional financial intermediaries, BigTech platforms accumulate data from interactions within their digital ecosystems, including e-commerce transactions, payments, and other online activities. These data can provide valuable information about borrowers' creditworthiness, allowing platforms to enter lending markets without relying exclusively on traditional credit histories or costly borrower screening. Moreover, as digital ecosystems expand, the information generated by their users can become increasingly valuable through economies of scale and data externalities (Boissay et al., 2021; Bergemann et al., 2022). The resulting informational advantages raise important questions about how platform entry changes competition with incumbent banks and the allocation of credit across lenders.

An important concern is that better-informed entrants may attract disproportionately creditworthy borrowers, leaving incumbent banks with a riskier portfolio. However, this effect is not mechanical. Banks can adjust their lending rates and invest in screening to respond to competition, while borrowers' preferences for different lending channels influence which loans are ultimately originated by each intermediary. Consequently, the impact of platform entry on bank portfolio quality depends on the interaction between information, pricing, screening, and borrower selection. This paper asks how competition from a data-rich digital platform affects an incumbent bank's optimal lending decisions and the risk composition of its loan portfolio. In particular, we examine when platform entry worsens the bank's portfolio quality, how the bank responds through interest rates and screening, and whether these changes imply a deterioration in the efficiency of credit allocation.

To address these questions, we develop a sequential model of credit-market competition

between an incumbent bank and a digital platform. Borrowers differ in their underlying credit risk and in their relative convenience of borrowing from either lender. The bank can serve the entire market and chooses a costly screening intensity to improve its assessment of borrowers' repayment probabilities. By contrast, the platform possesses a screening technology based on information accumulated through previous activities within its ecosystem. A fraction of borrowers has already adopted the platform, and only these borrowers can obtain platform credit. Importantly, adoption also determines the precision of the platform's signal: as the ecosystem expands, the platform becomes better able to identify borrowers' creditworthiness without undertaking an additional lending-specific screening investment.

The interaction between market reach and information quality is central to our analysis. Greater platform adoption simultaneously increases the population exposed to competition and improves the entrant's ability to distinguish between safe and risky borrowers. We take prior ecosystem adoption as exogenous and independent of borrower creditworthiness, isolating how information generated in other markets affects subsequent lending competition. The bank first commits to its screening intensity, a uniform loan rate, and a signal-contingent lending policy. The platform observes these choices and decides whether and on what terms to enter. Borrowers then receive offers conditional on lenders' screening outcomes and select their preferred acceptable contract, or remain unfunded. Horizontal differentiation in lending convenience determines both borrowers' participation decisions and the allocation of loans when competing offers are available.

Our first result identifies a mechanism through which platform competition can increase the riskiness of the bank's retained portfolio. At unchanged lending decisions, entry raises the bank's expected default frequency whenever the platform displaces a larger proportion of safe than risky bank loans. In the benchmark competition region, safe borrowers can receive mutually acceptable offers from both lenders, whereas risky borrowers do not have the same degree of overlap. Consequently, the platform primarily removes profitable safe lending opportunities from the bank. The bank's portfolio can therefore become riskier even

when the mass of risky loans it originates does not increase. This distinction is important: a deterioration in portfolio quality need not reflect additional risk-taking by the incumbent, but can instead arise from the selective departure of safer borrowers.

Our second result concerns the incumbent's endogenous response to this competitive pressure. Although the platform attracts some of the bank's profitable customers, the bank does not necessarily respond by lowering its interest rate. In the interior benchmark competition region, we show that its optimal rate, conditional on screening, exceeds the corresponding monopoly rate. The mechanism follows from the bank's inability to target discounts exclusively to contestable borrowers. A lower common rate can help recover some customers from the platform, but it also reduces revenues on loans the bank would have retained and can attract additional risky applicants. The bank therefore balances the benefits of defending market share against the costs of repricing its entire loan portfolio. Under the benchmark conditions, the latter considerations lead to a higher optimal rate.

The response of screening is more nuanced. Greater screening accuracy allows the bank to identify profitable safe borrowers and avoid financing unprofitable risky projects. Platform entry weakens the first benefit because some identified safe borrowers will ultimately choose the competing lender. At the same time, screening continues to protect the bank from expected losses on risky loans, while changes in the platform's pricing also affect the return to information acquisition. In the benchmark equilibrium, these opposing forces generate a relatively moderate screening adjustment compared with the bank's pricing and portfolio-composition responses. Thus, the deterioration in bank portfolio quality is driven primarily by the reallocation of borrowers across lenders rather than by a substantial reduction in the incumbent's screening effort.

We then distinguish the effects of expanding platform adoption from improvements in its information-processing technology. Holding adoption fixed, more accurate platform information increases the likelihood that the entrant identifies and attracts safe borrowers while reducing its willingness to finance risky ones. This strengthens adverse selection in the bank's

retained portfolio and reduces the returns from identifying profitable borrowers. Our numerical equilibrium analysis confirms that more effective platform information is associated with higher bank portfolio default frequencies and lower bank profits, although the response of screening is comparatively limited. We also examine differences in lenders' funding costs. These affect both lending margins and the extent to which borrowers find competing offers acceptable. In particular, tighter funding conditions can change the market-coverage regime, altering the equilibrium responses of lending rates and screening. Hence, the effects of informational competition depend not only on relative screening technologies but also on the conditions under which lenders compete for borrowers.

Finally, we examine the implications for aggregate credit allocation and productive efficiency. A deterioration in the bank's portfolio quality does not necessarily imply that platform entry makes lending less efficient. In our framework, risky projects have negative expected net returns, while safe projects generate positive productive surplus. Platform entry can finance safe borrowers who would otherwise remain excluded from credit, even as it diverts some profitable lending toward an intermediary with higher funding costs. These forces generate a trade-off between credit creation, inefficient duplication or reallocation of funding resources, and the costs of screening errors. In the benchmark numerical equilibrium, productive efficiency initially declines relative to bank-only lending but subsequently improves as platform adoption expands and additional safe projects receive financing. Therefore, bank portfolio risk and aggregate productive efficiency need not move in the same direction.

Our analysis contributes to the literature on information acquisition and credit-market competition. Hauswald and Marquez (2003, 2006) establish the strategic role of information in lending, while more recent contributions study how advances in information technology reshape lender competition and credit allocation (Vives and Ye, 2025a,b; He et al., 2024). We complement these analyses by considering an entrant whose information is generated outside the lending market and whose screening precision improves with the size of its pre-existing ecosystem. Our framework also relates to research on platform-generated information and

data externalities (Parlour et al., 2022; Bergemann et al., 2022; Chiu and Koeppl, 2026), as well as work examining the implications of platform lending for credit-market competition and welfare (Biancini and Verdier, 2023; Gambacorta et al., 2023; Li and Pegoraro, 2026). Our distinctive focus is on how the interaction between platform reach, information quality, and endogenous incumbent responses determines the composition of loans retained by traditional banks.

The model highlights why the consequences of BigTech lending cannot be assessed solely through changes in lending volumes, interest rates, or the default frequency of incumbent portfolios. Understanding these effects requires distinguishing borrower selection into credit applications from the allocation of originated loans, and recognizing that improved information at one intermediary can alter the risk composition of its competitors. This distinction also provides an empirical implication: greater platform penetration should disproportionately displace safer bank lending opportunities where platform-generated data are more informative about borrower creditworthiness.

The remainder of the paper is organized as follows. Section 2 discusses the related literature. Section 3 presents the model. Section 4 characterizes the bank-only benchmark, and Section 5 analyzes competition with the platform. Section 6 conducts some comparative statics on our equilibrium, while Section 7 studies the implications for productive efficiency and welfare. Section 8 concludes and sets some venues for future research.

2 Related literature

Information acquisition and credit-market competition. Information affects lending competition through both borrowers' selection and lenders' incentives to acquire information about their prospective customers. Hauswald and Marquez (2003, 2006) examine the strategic value of information in credit markets; Marquez (2002) studies how information dispersion shapes credit quality and lending rates, while Broecker (1990) highlights the se-

lection problems created by competing creditworthiness tests. More recently, Vives and Ye (2025a,b) show how advances in monitoring technology and technological entry alter lending competition, rates, and the allocation of credit. Also, He et al. (2024) distinguish information breadth from signal precision through explicitly discussing the differences between hard and soft information. We retain endogenous costly bank screening, but make the entrant's signal to depend on information generated through activities other than lending. We assume the quality of this signal improves with ecosystem adoption. This distinction makes the composition of the bank's retained loans a central equilibrium outcome.

Data externalities and platform-generated information. Digital platforms accumulate information through activities that need not involve credit. Parlour et al. (2022) study how competition for payment flows changes lending information, while Chiu and Koeppl (2026) analyze the data-network-activities channel of platform finance. Bergemann et al. (2022) establish how data on one user can be informative about others. These contributions motivate an adoption-dependent signal technology: as the platform's ecosystem expands, its ability to screen borrowers improves even without an additional lending-specific screening investment. Unlike models of data collection or sharing, we take prior adoption as given and concentrate on its consequences for credit-market competition.

Platform lending and credit allocation. Platform lenders can change credit provision through mechanisms other than conventional price competition. Li and Pegoraro (2026) emphasize enforcement through access to an e-commerce ecosystem, Gambacorta et al. (2023) study the interaction between marketplace fees and credit terms, and Biancini and Verdier (2023) examine bank–platform competition and credit expansion. Evidence on the lending technology itself is provided by Liu et al. (2022), who document that BigTech loans differ from traditional bank loans in size, maturity, and rates. Our model abstracts from network enforcement, cross-subsidization, and maturity choice. It instead isolates how a platform's expanding reach and improving credit signal jointly affect the bank's pricing, costly screen-

ing, and the relative displacement of safe and risky loans.

3 Model

Consider an economy with a continuum of mass one of households $i \in \mathcal{I}$, an incumbent monopolist bank, and one Bigtech platform that is a potential entrant in the credit market. We refer to intermediaries with a subindex $\ell \in \{b, p\}$ for the bank and platform, respectively.

We set our attention in a one-stage sequential game where households are endowed with a productive project. We assume households are penniless, so they cannot self financing their productive investment. For finding financing they seek a loan from at most two intermediaries: the bank or an entrant digital platform.

Households are heterogeneous along two dimensions. The first is their relative riskiness, $j \in \{S, R\}$, which impacts the success probability of their project, θ^j. The other dimension is their preferences for each lender, x_i.

We assume that the economy has an exogenous share μ of these households that are platform adopters. This can be thought as the population that has interacted in the platform's ecosystem in the past. Namely, if they have conducted any transactions in the platform's marketplace, used one of its products or shared information actively in it. Importantly, the location x_i of the borrower in the lender-characteristic space is assumed to not be correlated with platform adoption.

This is relevant because it controls the share of borrowers that are contestable by the platform, given the following assumption:

Assumption 1 (Exogenous Market Segmentation)**.** *Bigtech platform can only lend to households that have previously adopted the platform*

Both intermediaries use imperfect screening signals to infer a borrower's repayment probability. The bank chooses costly screening intensity I, whereas the platform has access to a costless signal whose informativeness improves with μ. The signals are conditionally inde-

pendent given the entrepreneur's true credit type. Lenders share the same prior distribution of credit types because our stated assumption that previous platform adoption and location are independent of creditworthiness. Intermediaries select a single rate to charge to successful applicants (i.e. those considered credit-worthy given the expected return for the lender, conditional on the signal). For the bank we denote this rate as R_b, whereas the platform chooses an interest rate R_p

Given these elements, we will have a game in which the bank chooses its screening intensity, a single posted gross repayment R_b, and a signal-contingent loan-offer policy $\mathcal{M}_b \subseteq \{s, r\}$. The platform observes this commitment and decides whether to enter by selecting its own posted repayment R_p and offer policy $\mathcal{M}_p \subseteq \{s, r\}$ (the empty set represents no entry). Once the screening signals are realized, contracts are offered automatically according to the committed policies, and contingent on the result of the signal. Borrowers then choose at most one available contract or decline financing. After this, projects are realized and successful projects repay the accorded amount to lenders. The timing is described in detail below.

3.1 Timing

At the beginning of the period each entrepreneur privately observes their credit type and location before lenders choose their actions. Both intermediaries know the joint distribution of these characteristics, the aggregate adoption share, and the screening technologies. The bank does not observe whether a household has adopted the platform; the platform knows however who its own users are.

We assume also that the share of previous platform adopters, μ, is given. This adoption has nothing to do with previous lending activities, since the platform is a prospective entrant. Rather, we assume this adoption has more to do with having conducted activities within a platform's ecosystem or used one of their products. This will constitute the contestable segment, since by assumption 1 the complementary share is outside the reach of the platform. We see this as an assumption that accommodates the fact that lending is a complementary

activity to the platform's main business source. Hence, lending is not a main reason to have joined the digital platform in the first place.

As mentioned, this is a sequential game in the credit market, where an incumbent (i.e the bank) will face the threat of competition by a potential Bigtech platform entrant. In our economy the bank has available two instruments to compete in the credit market: choosing the intensity with which to screen applicants[1] , and the interest rate it will offer the applicants it accepts.

Both intermediaries can infer the borrower's credit type through a screening technology. However, they are heterogeneous on how their technologies are. The bank has access to a technology that requires costly effort, I, to operate. On the other hand, we assume the platform will have access to a costless signal that will be a function of the mass of adopters, μ. This structure relates to the fact that Bigtech companies and digital platforms can scale their users' data to infer valuable characteristics about them through their interactions in its ecosystem.

bank will choose how intensively it would screen at the beginning of the game, before posting the interest rate it will charge to successful applicants. The platform observes the screening effort and the rate committed by the bank and then decides whether to enter into the credit market or not, and in which segment to do so. Given credit types are binary, the decision for each intermediary will be reduced to decide if they offer a contract after observing only a particular signal, $m = \{s, r\}$ for the bank and $n = \{s, r\}$ for the platform, or regardless of it. Thus the screening signal is reduced to a test in which the intermediary, given the expected profitability of the identified borrower, decides to deliver an offer or not.

After both lenders' rates are known, screening signals realize and contracts to prospective borrowers are offered. We assume each intermediary offers a single take-it-or-leave-it contract to a successful applicant. Whether a contract is offered is conditional on the result of the screening signal.

[1]This can be also thought of how much to invest into a screening technology. This is a standard approach based on the seminal papers by Hauswald and Marquez (2003; 2006)

Finally, borrowers will pick the contract one that leaves them with highest expected utility. Note that given their location shock and the available offers, some of them may decide to stay out of credit. After financing is granted, projects are realized and repayment is done to the corresponding intermediary if it was successful.

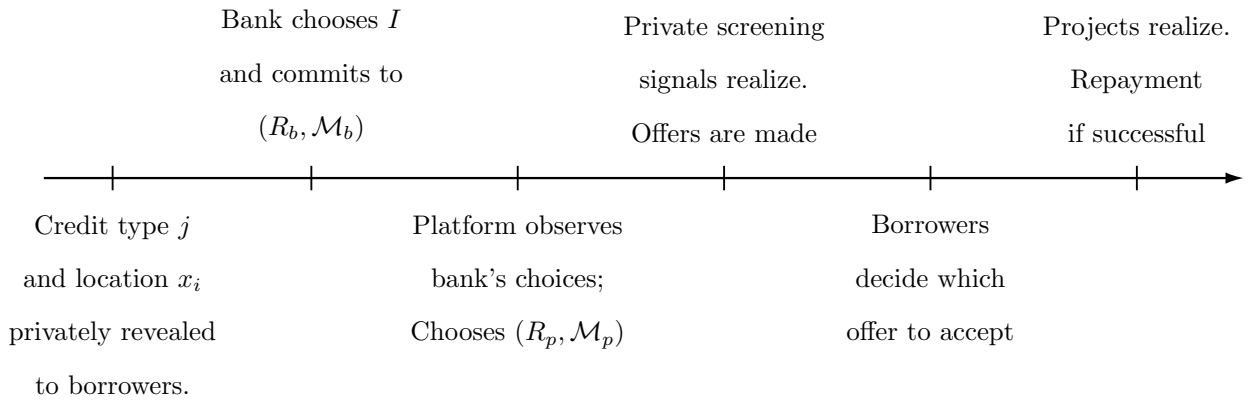


Figure 1: Timing of the sequential bank–platform lending game.

3.2 Agents

3.2.1 Households

There is a continuum of mass 1 of households, which are assumed to be risk-neutral. As mentioned, households are heterogeneous on their repayment probability, θ^j, and in their location, or distance, x_i to a lender. These are private information of the household and revealed to them at the beginning of the game.

The θ^j of a household is important in the sense that it will influence the *probability of success* of their productive project once financing is granted. Note that these names of types are used to express their *relative riskiness*, as both types of households will have some probability that their projects do not succeed.

We assume the distribution of credit types is binary , such that there are only two credit

types in the economy $\theta \in \{\theta^S, \theta^R\}$ where:

$$\theta^j = \begin{cases} \theta^S \text{ with probability } p \\ \theta^R \text{ with probability } 1-p \end{cases}$$

In what follows I will call those households of type $\theta^j = \theta^S$ a *safe-type*. Conversely, those with $\theta^j = \theta^R$ will be referred to as *risky-type*. We assume that $\theta^S > \theta^R$, so that safe-type borrowers have higher repayment probability than risky-types.

The repayment probability of the household is relevant for intermediaries' financing decisions as the eventual success (and return) of the investment project is depends on it.

The households are endowed with a productive project that needs one unit of financing to operate. The project's output, $V^j(\theta)$, is stochastic and contingent on the credit type $j \in \{S, R\}$, such that:

$$V^j(\theta^j) = \begin{cases} \frac{Y^j}{\theta^j} \text{ with probability } \theta^j \\ 0 \text{ with probability } 1-\theta^j \end{cases}$$

This is a sufficiently general specification so we have that productive project's output is allowed to have different expected value per credit type. Particularly, we could have combinations in which the risky-types could have higher base returns. The expected returns for each type will be the main concern for lenders in deciding whether to grant a contract offer or not depending on the *perceived* credit type according to their screening signal. Note that under this specification we can refer to Y^j as the expected output for a borrower of credit type j.

Particularly, we will apply the following assumption for the remaining of the paper concerning the profitability of each type of borrower from the lenders' point of view:

Assumption 2. *Calling c_ℓ the unit cost of lending for each lender, we will have that a funded*

true risky project has negative net present value for both lenders, such that:

$$Y^R - c_\ell < 0 \quad \textit{for } \ell = \{b, p\}$$

This does not mean that a true risky borrower will never receive funding or that it is not profitable for her to do so. A risky borrower might get funding if she is misclassified as safe by either intermediary, or if a particular lender finds it profitable to offer a contract under either signal.

In addition to differences in creditworthiness, borrowers are heterogeneous in their compatibility with traditional and digital lending channels. Such compatibility may reflect familiarity with lending procedures, the convenience of loan origination and servicing, documentation requirements, or preferences for relationship-based versus platform-integrated financial services.

Following the spatial approach to banking competition (Vives and Ye, 2025a), we represent this heterogeneity as a linear distance. To be precise, the incumbent bank and the digital platform are located at opposite endpoints of a unit interval, while borrowers are uniformly distributed according to a privately observed characteristic x_i. A borrower at location x_i incurs non-pecuniary costs tx_i when borrowing from the bank and $t(1-x_i)$ when borrowing from the platform, where $t > 0$ measures the importance of lender differentiation.

These costs are incurred conditional on obtaining financing and are independent of project creditworthiness. Accordingly, the borrower's location affects both the relative attractiveness of competing loan offers and the decision to undertake a project when only one lender makes an offer. Borrowers may optimally decline financing when the expected surplus from operating the project is insufficient to compensate for the repayment obligation and the non-pecuniary costs associated with the available offer.

This location is predetermined and privately known to borrowers, and it is private information for them. Thus, from the point of view of the lenders x_i is a random variable whose

particular realization they do not observe, although they know their distribution. Particularly we assume that a location for a borrower i is drawn from an uniform distribution such that:

$$x_i \sim \mathbf{U}[0,1]$$

We impose an exogenous share μ of the households to be previous platform adopters. As we explained above, we take this fraction as being those households that have consumed in a digital platform's marketplace, conducted some activities on its ecosystem, or used one of their products extensively. Since the main entrant we have in mind is a Bigtech, these are companies that have an important presence in a myriad of markets. Whether it is on software, cloud computing or hardware, all of these companies operate a business focused on technology, not financial products *per se*.

Having this fraction is important to highlight the role that data externalities and network effects (in a reduced form) have in our model. Particularly, how easy is for a Bigtech entrant to scale up their market power and presence in other businesses to the credit market. From the household's side, being a platform adopter is important since it conditions its prospective borrowing opportunities, as per assumption 1.

The remaining fraction $1-\mu$ are assumed to belong to the non-platform-adopters. Those are the potential captive users for the bank, since they have no outside option in the credit market.

Importantly, the credit-type, the location and adoption status (whether the borrower has adopted the platform's ecosystem previously or not), are mutually independent. This

Borrowers, after learning privately their location in the unit line and their credit type, will apply for credit to the intermediaries that are available to them. They will get an offer from the intermediary ℓ whenever, after a screening signal is realized, they find it profitable to lend to the borrower based on the result of the screening signal.

We assume the screening signal is imperfect. Hence, we need to differentiate the true type j of the borrower with the perceived type for a given lender, which is based on the screening signal. Particularly, let $m \in \{s, r\}$ for the bank and $n \in \{s, r\}$ for the platform denote the outcome of the screening signals.

We purposely use lowercase letters for the *perceived* type for an intermediary given its resulting signal, while we keep capital letters for the borrower's *true* type. For example, a borrower $\theta = \theta^S$ is a *true* safe-type borrower. However, they might be perceived as a risky borrower by the bank ($m = r$) be cut off out of credit.

Conditional on receiving an offer, a borrower's expected utility from each intermediaries' contract is:

$$u_b(\theta^j) = \theta^j \left(\frac{Y^j}{\theta^j} - R_b \right) - t x_i \tag{3.1}$$

$$u_p(\theta^j) = \theta^j \left(\frac{Y^j}{\theta^j} - R_p \right) - t(1 - x_i) \tag{3.2}$$

In the case that both contracts (one by the bank and one by the platform) are available to the borrower we will have that she will take the one that gives her higher expected utility. Then, she will take the bank's contract:

$$u_b(\theta^j) \geq u_p(\theta^j) \iff x_i \leq \underbrace{\frac{1}{2} + \frac{\theta^j (R_p - R_b)}{2t}}_{\bar{x}^j} \tag{3.3}$$

Thus, there exists a threshold location $\bar{x}^j$ in which a borrower of type j is indifferent between the two offers.

Note that condition (3.3) only applies for platform adopters that received offers from both lenders. It might well be the case that they have only one rate available, or that the borrower is not a platform adopter. In the latter case, she can have at most one prospective offer.

Whenever a borrower has available a unique offer from lender ℓ the previous condition needs to include only its reservation value, which is normalized here to 0. In general, we

can say that there is an additional outside option for the borrower, other than get credit and operate the product. To be concrete, let us refer to the utility linked to this option as $u_0 = v = 0$. We assume the outside option is the same across borrowers and independent of type.

Not that the situation we are describing arises when either the borrower is a non-platform adopter or whenever one of the available lenders rejected her credit application based on the resulting screening signal. In that case, a borrower will accept credit from the bank whenever having only its contract as a feasible choice as long as:

$$x_i \leq \frac{Y^j - \theta^j R_b}{t} \equiv \bar{x}_b^j \tag{3.4}$$

Similarly, whenever a borrower has only the platform's offer available, she will take it as long as it gives her higher expected utility than her outside option. Her threshold $\bar{x}_p$ is characterized as:

$$x_i > 1 - \frac{Y^j - \theta^j R_p}{t} \equiv \bar{x}_p^j \tag{3.5}$$

In what follows we will refer to market coverage. For concreteness we should introduce the following definition to be used over the explanations of our equilibrium:

Definition 1 (Market coverage)**.** *Conditional on both intermediaries extending loan offers, the market is covered for borrowers of type j whenever every realization of their relative lender preference admits at least one contract yielding nonnegative expected utility.*

This requires the bank's participation threshold to be weakly greater than the platform's, or equivalently, that the sum of the two lenders' type-specific expected repayment obligations does not exceed twice expected project output minus the total differentiation cost.

When the condition holds strictly and the participation thresholds are interior, a positive mass of borrowers finds both contracts acceptable. Their allocation is determined by the indifference threshold between lenders. Otherwise, there is an interval of borrowers who would decline financing even if both lenders extended offers. In this uncovered region, the condi-

tional demand facing each lender coincides with its demand when it is the only intermediary offering a loan.

Market coverage is defined conditional on the availability of both offers. It does not imply that all borrowers receive financing, since credit remains subject to lender eligibility and screening decisions.

Note that $\bar{x}^j = \frac{\bar{x}_b^j + \bar{x}_p^j}{2}$, thus lying in the middle point between the two one-offer thresholds. Given this, one final aspect to discuss is whether there might be borrowers that can decline credit even when two viable contracts are available. We have that this is the case whenever $\bar{x}_b^j < \bar{x}_p^j$. Note that the whole market will be served, if two offers are available, as long as $\bar{x}_p^j < \bar{x}^j \leq x_b^j$. In that case there are "no gaps" in the space of viable offers from the borrowers' point of view. In such a case, borrowers with location to the left of $\bar{x}^j$ take the bank's offer, while those to the right of this threshold prefer the platform's offer. Figure 2 below shows this graphically.

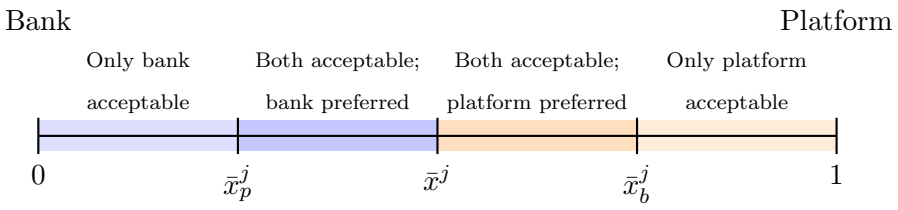


Figure 2: Thresholds whenever two offers are available. Intervals indicate the set of acceptable contracts.

This situation is more likely whenever interest rates are lower, expected output is higher or when horizontal differentiation is not too strong. This last element coincides with a low value of our parameter t, which dictates how different are both offers, from a convenience perspective, for a borrower.

On the other hand, if $\bar{x}_b^j < \bar{x}_p^j$ we could have the possibility that there is an uncovered or underserved fraction of the market even if borrowers have more than one offer to pick from. As shown in figure 3, borrowers in such an economy could have two offers with neither being

acceptable to them. This population would stay out of the credit market.

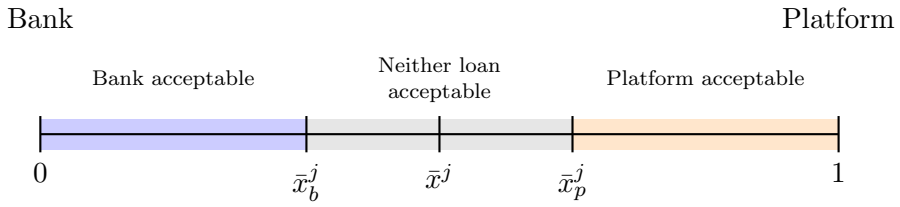


Figure 3: The three type-j cutoffs with an uncovered middle interval. The indifference cutoff lies within the interval where neither offer yields positive utility.

Hereafter we will focus our attention on a case like the one presented in figure 2. That is, that borrowers may always have a viable offer at hand if the screening signal is such that lenders decide to offer her credit.

Uniformly distributed locations allow us to express the shares of type-j borrowers accepting a lender's offer directly in terms of these thresholds. Let D_ℓ^j denote the conditional demand for lender ℓ when it is the only lender offering credit, and let $\widetilde{D}_\ell^j$ denote its conditional demand when both lenders extend offers.

$$\begin{aligned} D_b^j &= \bar{x}_b^j, & D_p^j &= 1 - \bar{x}_p^j \quad \text{(only one offer viable)} \\ \widetilde{D}_b^j &= \bar{x}^j & \widetilde{D}_p^j &= 1 - \bar{x}^j \quad \text{(both offers viable).} \end{aligned} \tag{3.6}$$

These quantities describe the allocation of type-j borrowers *conditional on the offers being available*; they do not yet incorporate the probabilities of passing screening or the share of platform adopters. Outside the considered case, participation constraints and truncation to the unit interval must be imposed when computing the corresponding demand shares. Thus, our focus on the case with $\bar{x}_p^j < \bar{x}_b^j$ allows for a cleaner description of the main effects delivered by our framework.

Applying to any lender is costless. Borrowers observe their credit types, locations, and the posted repayment rates before applying, but not the realizations of the lenders' screening signals. They may therefore apply to every eligible lender whose contract would give nonnegative expected utility if approved. In particular, a type-j platform adopter with $\bar{x}_p^j \leq x_i \leq \bar{x}_b^j$

may apply to both lenders, as either application may be rejected after screening. Intermediaries subsequently implement their previously committed signal-contingent approval policies, and borrowers choose their preferred acceptable offer among those received.

The participation thresholds also imply that the composition of loan applicants need not coincide with the population distribution of credit types. Because intermediaries know these thresholds when committing to their lending strategies, they anticipate this selection. The conditional demands above, combined with screening and approval outcomes, will determine the volume and credit-risk composition of loans originated by each intermediary.

3.2.2 Bank

There is a monopolist bank granting financing to producers conditional on the results of its screening activity. The bank acts as a leader-incumbent in the credit market.

To conduct its lending activities, the bank has access to a perfectly elastic supply of funds at a cost of c_b per unit lent. We can relate this to the interbank rate, or some reference rate in the economy. This distinction will be useful when interpreting some comparative statics later on.

The bank is endowed with a screening technology that allows it to infer the credit-type of the prospective borrowers. The bank can decide how much effort $I \in [0, 1]$ to put into the screening technology. This is chosen at the beginning of the game, although the actual signal is revealed after both intermediaries have set their rates and when prospective borrowers apply for credit.

We assume the cost of screening is a convex and increasing function of effort. Particularly, we assume a for that makes it infinitely costly for the bank to exert all the possible effort

$$c(I) = \kappa \frac{I^2}{1 - I}$$

Once screening signals are realized, an applicant receives an offer whenever the observed

signal belongs to the set of outcomes for which the bank has committed to extend credit. Thus, screening allows the bank to select borrowers for financing, while all approved borrowers face the same contractual interest rate. Since screening is imperfect, borrowers of different true credit types may receive identical offers, and lending under a given signal may be profitable even when some borrowers in that group have negative net present value projects.

To be precise, after choosing I and committing to its interest rate, the bank gets a signal, $m \in \{s, r\}$, about an applicant's type. Its precision, $\gamma(I)$, is a function of the effort put into screening activity. Formally:

$$Prob(m = s|\theta = \theta^S) = Prob(m = r|\theta = \theta^R) = \gamma(I)$$

We assume the accuracy of the signal is increasing in I. We adopt a specification similar to that of Hauswald and Marquez (2003) where:

$$\gamma(I) = \frac{1 + I}{2}$$

This means that we can write the unconditional probability of each one of the two signals as:

$$Prob(m = s) = \gamma(I)p + (1 - \gamma(I))(1 - p)$$
$$Prob(m = r) = \gamma(I)(1 - p) + (1 - \gamma(I))p$$

However, these probabilities require a refinement. Since borrowers privately observe their credit types and locations before applying for credit, the composition of the bank's applicant pool need not coincide with the distribution of credit types in the overall population. Recall from the households' problem that the conditional demands for the bank's contract, D_b^j, also represents the probability that a type-j borrower applies to the bank.

Accordingly, the bank anticipates the selection of borrowers into applications when choosing its lending terms. Denoting by p_b the probability that a randomly selected bank applicant is of the safe type, this prior is determined by the relative participation probabilities of safe and risky borrowers. Hence, the relevant prior that an applicant is a true safe type before a screening signal turns into the following:

$$p_b(R_b) = \frac{pD_b^S(R_b)}{pD_b^S(R_b) + (1-p)D_b^R(R_b)} \tag{3.7}$$

Upon receiving the screening signal, the bank updates its prior about a borrower's credit type. Let β^S denote the posterior probability that an applicant receiving a favorable signal is truly safe, while β^R denotes the posterior probability that an applicant receiving an unfavorable signal is truly risky. Both posterior beliefs depend on screening intensity and on the bank's posted interest rate, since the latter affects the composition of applicants:

$$\beta^S(I, R_b) = \frac{\gamma(I)p_b}{\gamma(I)p_b(R_b) + (1-\gamma(I))(1-p_b(R_b))}$$

$$\beta^R(I, R_b) = \frac{\gamma(I)(1-p_b(R_b))}{\gamma(I)(1-p_b(R_b)) + (1-\gamma(I))p_b(R_b)}$$

Both these posteriors are increasing in $\gamma(I)$, and satisfy that whenever $I = 0$ the posterior stays at the prior, p_b. These beliefs characterize the information conveyed by screening, but they do not necessarily describe the composition of loans ultimately originated by the bank. Some approved borrowers may also receive an offer from the platform and choose to obtain financing there. This additional selection is captured by the conditional demand when both intermediaries offer credit.

Importantly, the bank commits to its interest rate and signal-contingent approval policy before applications and screening outcomes are observed. Thus, its decisions incorporate the anticipated composition of applicants ex ante, while its choices cannot be revised after screening signals are realized.

3.2.3 Platform

We consider a single BigTech platform that may enter the credit market and compete with the incumbent bank. As described in the households' problem, an exogenous fraction $\mu \in [0, 1]$ of borrowers has previously used the platform's non-financial services. These activities generate information that the platform can exploit when evaluating credit applications. The platform can extend loans only to previous adopters, whereas the bank may lend to the entire population. Thus, μ determines both the platform's potential customer base and the scale of its pre-existing informational resources.

The platform is endowed with a credit-screening technology based on data gathered through its ecosystem. Unlike the bank, it does not choose a costly screening investment, as the data is gathered as a byproduct of its users' activity. In this case, its signal precision is determined by the adoption rate and the effectiveness with which the platform uses its data. Let $n \in \{s, r\}$ denote the platform's signal, and let $\delta(\mu; \alpha)$ denote its accuracy, then we can define the latter as:

$$\delta(\mu; \alpha) = \frac{1 + \mu^{1/\alpha}}{2}, \quad \alpha \geq 1. \tag{3.8}$$

As with the bank's problem, we assume the probability of an accurate signal is symmetric across types:

$$\Pr(n = s \mid \theta = \theta^S) = \Pr(n = r \mid \theta = \theta^R) = \delta(\mu; \alpha)$$

Conditional on the borrower's true credit type, the platform's signal is independent of the bank's screening signal and of borrower relative lender preferences.

The specification implies that $\delta(0; \alpha) = \frac{1}{2}$ and $\delta(1; \alpha) = 1$. Thus, without an established ecosystem the signal is uninformative, while larger adoption improves its accuracy. For $0 < \mu < 1$, a higher α increases precision at a given adoption level. When $\alpha > 1$, accuracy is increasing and concave in adoption: the platform realizes relatively large informational gains at low levels of adoption, with diminishing marginal improvements as the ecosystem grows. This reduced-form relationship captures the possibility that data from one user are useful

for predicting the characteristics of other users, as emphasized by Bergemann et al. (2022). It does not require each additional adopter to generate increasing marginal informational returns.

This is also motivated by the fact that digital platforms usually employ data which is generated from non-financial activities by its users. In Chiu and Koeppl (2026) the focus is on how this data can be scaled up to improve payments. We adopt this view to look at the effects on the credit market. Note that we do not make a distinction between hard and soft information (He et al., 2024), rather we assume the data that the platform gathers about a borrower is always useful regardless of type in identifying a borrower's credit type.

In our model, the platform acts after observing the bank's committed screening intensity, interest rate, and approval policy. Then it will make the decision to enter into credit provision by choosing a common interest rate R_p and a set $\mathcal{M}_p \subseteq \{s, r\}$ of screening outcomes for which credit will be offered. Thus, an applicant receiving a signal in $\mathcal{M}_p$ is offered the same rate R_p, regardless of its true credit type. Choosing $\mathcal{M}_p = \varnothing$ corresponds to not lending. As with the bank, these choices are made before applications and screening signals are observed, and the platform subsequently implements its committed approval policy.

We denote the platform's unit funding cost of lending by c_p. In our benchmark, the bank has a funding advantage, which may be expressed as $c_p = \psi c_b$ with $\psi > 1$. This captures the possibility that the incumbent benefits from established deposit relationships and other comparatively inexpensive sources of funds.

Because applications are submitted after interest rates are announced, the platform's applicant pool may differ in credit-type composition from its eligible population. Recall that D_p^j is the fraction of type-j borrowers willing to accept a platform loan when it is the only available offer, as per expression (3.6). Given our assumptions, D_p^j also gives the type-j application probability among previous platform adopters. Accordingly, when the applicant

mass is positive, the prior probability that a platform applicant is safe is:

$$p_p = \frac{pD_p^S}{pD_p^S + (1-p)D_p^R}.$$

After observing its signal, the platform updates this prior in a Bayesian way. Similarly as for the bank, the probabilities that an applicant is truly safe following a favorable signal, or truly risky following an unfavorable signal, are respectively:

$$\beta_p^S = \frac{p_p\delta(\mu;\alpha)}{p_p\delta(\mu;\alpha) + (1-p_p)[1-\delta(\mu;\alpha)]},$$

$$\beta_p^R = \frac{(1-p_p)\delta(\mu;\alpha)}{(1-p_p)\delta(\mu;\alpha) + p_p[1-\delta(\mu;\alpha)]}.$$

These do not determine directly the ultimate composition of the platform's loan portfolio. Note that some approved applicants may also receive a bank offer and choose to borrow there. The distinction is captured by the conditional demand $\widetilde{D}_p^j$, for the case when a borrower has two prospective viable offers.

The platform therefore anticipates two forms of selection when setting its interest rate and approval policy: selection into applications, which depends on D_p^j, and selection between lenders after approval, which also depends on $\widetilde{D}_p^j$. Together with the bank's screening decisions, these will determine the platform's expected lending.

4 Benchmark with only bank lending

Let us consider now a benchmark of an economy where the incumbent bank is the only available lender. This example would helps us characterize how loan pricing and costly screening jointly determine the composition of originated credit. The bank will then choose its common rate to successful applicants, screening intensity, and approval policy in order to

maximize its expected profits. We focus on lending after favorable signals, which is optimal for the benchmark parameters; while we conduct the relevant comparisons with the other policies and participation boundaries are in Appendix A.4.

4.1 How does the bank price and screen its applicants?

Under favorable-signal lending, the masses of safe and risky loans are

$$L_0^S = pD_b^S\gamma(I), \qquad L_0^R = (1-p)D_b^R[1-\gamma(I)]. \tag{4.1}$$

Each mass is the product of the true-type share, the probability of applying, and the corresponding approval probability. Hence, with $m_b^j \equiv \theta^j R_b - c_b$, the bank's ex ante expected profit is

$$\Pi_b^M(R_b, I) = L_0^S m_b^S + L_0^R m_b^R - C(I). \tag{4.2}$$

Under the maintained project-return assumption, risky loans have a negative expected margin whenever a risky borrower is willing to borrow. Consequently, the bank benefits both from identifying safe applicants and from rejecting risky ones.

Whenever we have an interior solution for the interest rate its value will be unique. Let us call R_b^M to the optimal interior monopoly rate for the bank, which we can characterize through the following first order condition:

$$R_b^M(I) = \frac{1}{2}\left[\frac{p\gamma\theta^S(Y^S + c_b) + (1-p)(1-\gamma)\theta^R(Y^R + c_b)}{p\gamma(\theta^S)^2 + (1-p)(1-\gamma)(\theta^R)^2}\right] \tag{4.3}$$

This rate balances larger expected repayments on retained loans against the reduction in applications. We can see that (4.3) tells us the optimal rate is a weighted average of the rates that would be optimal for each type in isolation, $\frac{(Y^j + c_b)}{2\theta^j}$, with weights reflecting population shares, favorable-signal approvals, and repayment probabilities.

Note that a more accurate screening signal gives greater weight to safe borrowers, since it

lowers the conditional rate whenever $\frac{(Y^S+c_b)}{\theta^S} < \frac{(Y^R+c_b)}{\theta^R}$. Higher funding costs would naturally raise the conditional rate for a fixed screening intensity.[2]

Simultaneously to choosing its common rate, the bank chooses optimally how much effort or investment to put into the screening technology. The bank will balance the cost of getting a more precise signal with the marginal returns of a better identification of borrower types:

$$C'(I) = \underbrace{\frac{1}{2}\left[pD_b^S m_b^S - (1-p)D_b^R m_b^R\right]}_{\Lambda(R_b)}. \tag{4.4}$$

The first order condition on screening in (4.4) shows that improved accuracy has two benefits: it finances more profitable safe applicants and prevents more loss-making risky loans. The marginal benefit $\Lambda(R_b)$ does not depend directly on I at a given rate, while $C'(I)$ is strictly increasing. Thus the bank's conditional screening choice is unique, and more costly screening lowers it when the rate is held fixed. The rate and screening conditions jointly characterize an interior candidate, while the bank's global policy choice is verified in the appendix.

4.2 What determines the risk of originated loans?

The composition of the applicant pool need not coincide with that of bank credit. Screening shifts this composition because a truly risky applicant is less likely than a safe applicant to obtain a favorable signal. The relationship is especially transparent in the odds of originating a risky rather than a safe loan:

$$\frac{L_0^R}{L_0^S} = \frac{1-p}{p}\,\frac{D_b^R}{D_b^S}\,\frac{1-I}{1+I}. \tag{4.5}$$

The first two terms describe the composition of applications; the last one captures the relative chances a risky applicant is funded due to changes in screening. The bank's expected default

[2]Appendix A.5 derives this expression and treats participation boundaries for a more complete characterization of the solution.

frequency is

$$\Omega_b^M = (1 - \theta^S) + (\theta^S - \theta^R)\frac{L_0^R}{L_0^S + L_0^R}. \tag{4.6}$$

Lemma 1 (Screening and bank portfolio composition)**.** *At a given rate, more accurate screening strictly reduces the risky share of originated loans. When screening is informative, that share is also below the risky share among bank applicants.*

The proof is in Appendix A.3. The lemma isolates the role of information from the selection already generated by participation. Figure 4 illustrates the distinction: applicants include many risky borrowers who do not receive bank credit, and the composition of originated loans is consequently much safer than the composition of applications. In equilibrium, changes in the bank's rate also alter the two application probabilities in (4.5); the direct screening result therefore need not describe every adjustment of the fully optimized portfolio.

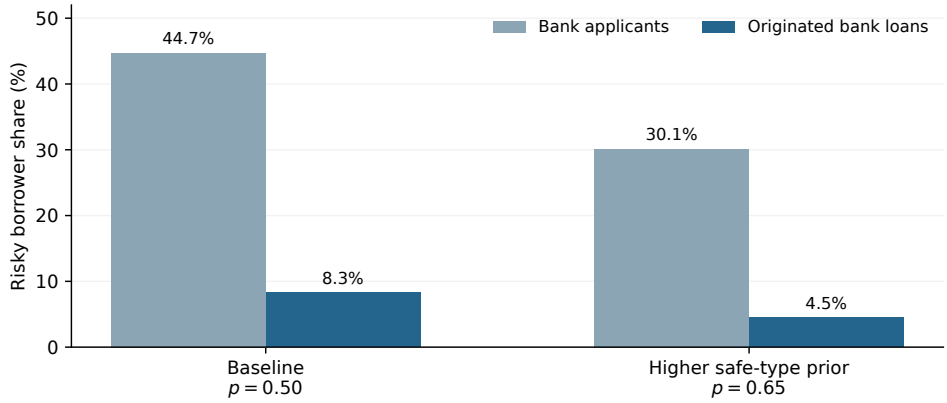


Figure 4: **Screening and loan composition.** Risky borrowers as a share of bank applicants and originated loans in the benchmark and in the higher-safe-prior calibration ($p = 0.50$ and $p = 0.65$, respectively; other parameters held fixed).

The bank-only benchmark therefore distinguishes two forms of selection: participation determines who seeks credit, and screening determines who receives it. With platform entry, approved borrowers may additionally choose a competing lender. The next section examines

how this additional allocation margin changes bank pricing, screening, and retained portfolio quality.

5 Competition with entrant digital platform

The bank-only benchmark establishes how borrower participation and screening determine credit allocation. Platform entry adds a second source of selection: borrowers whom the bank would finance may instead borrow from a rival with information generated outside the credit market. We examine first which loans the entrant displaces, then how the bank responds through pricing and screening. The distinction between displacement at given lending terms and the resulting equilibrium allocation is central to the effects on portfolio risk.

5.1 Credit allocation and equilibrium

Under favorable-signal lending, the bank originates

$$\begin{aligned} L_b^S &= p\gamma\left[(1-\mu\delta)D_b^S + \mu\delta\widetilde{D}_b^S\right], \\ L_b^R &= (1-p)(1-\gamma)\left[(1-\mu(1-\delta))D_b^R + \mu(1-\delta)\widetilde{D}_b^R\right]. \end{aligned} \tag{5.1}$$

The platform's originated loans are

$$\begin{aligned} L_p^S &= \mu p\delta\left[(1-\gamma)D_p^S + \gamma\widetilde{D}_p^S\right], \\ L_p^R &= \mu(1-p)(1-\delta)\left[\gamma D_p^R + (1-\gamma)\widetilde{D}_p^R\right]. \end{aligned} \tag{5.2}$$

Each mass incorporates the borrower's true type, the relevant approval probabilities and the choice of lender. In particular, a favorable signal is not equivalent to a safe loan. For either intermediary, expected profit is the sum of type-specific originated loans times their expected margins, $m_\ell^j \equiv \theta^j R_\ell - c_\ell$, less the bank's screening cost $C(I)$ when applicable. The general loan masses and profits for every approval policy are derived in Appendix B.

Equilibrium. For each commitment (R_b, I, M_b), the platform chooses the rate and approval policy that maximize its ex ante profit, including the option not to lend. The bank chooses its commitment anticipating this response. A competitive equilibrium consists of these optimal sequential choices, together with the borrowers' application and contract choices described in Section 3; in the event of multiple platform best responses, a specified selection applies. Both intermediaries optimally approve favorable signals in the benchmark, as verified against alternative policies and nonentry in Appendix B.

Benchmark competition region. At the benchmark equilibrium rates, safe borrowers have overlapping acceptable offers, whereas risky borrowers do not. The pricing and screening conditions below concern the interior of this region; the conditional monopoly and competitive rate optima used in the pricing comparison satisfy the same interior restrictions. The platform rate responds positively to the bank rate within this region. Appendix B.6 covers participation boundaries and other pricing regions. The loan-displacement result, by contrast, does not depend on the benchmark's particular market coverage.

5.2 Which loans does the platform displace?

Consider the bank-only masses L_0^S and L_0^R evaluated at the same bank rate and screening intensity as under entry. Define $\Delta_j = D_b^j - \widetilde{D}_b^j$, the reduction in type-$j$ bank demand when a competing offer is available. The bank-loan expressions in (5.1) show that the platform displaces $\mu p \gamma \delta \Delta_S$ safe loans and $\mu(1-p)(1-\gamma)(1-\delta)\Delta_R$ risky loans. Thus its effect on bank credit quality depends on the *relative*, rather than absolute, displacement of the two types. For this comparison, assume positive adoption and positive bank originations of both types, with favorable-signal approval by both lenders.

Proposition 1 (Relative displacement and portfolio risk)**.** *At unchanged bank decisions, platform entry increases the bank's expected default frequency precisely when it displaces a*

larger proportion of safe than risky loans:

$$\delta \frac{\Delta_S}{D_b^S} > (1-\delta)\frac{\Delta_R}{D_b^R}.$$

The proposition isolates selection through competing offers. In the benchmark, risky borrowers do not have mutually acceptable offers, so $\Delta_R = 0$: the platform removes safe bank lending opportunities without directly displacing risky loans. Yet uncovered risky demand is not essential. When both types are contested, the bank's loan quality deteriorates whenever safe loans are lost at the higher proportional rate. The complete accounting is in Appendix B.10.

Better platform information strengthens this mechanism. At given lending terms, greater precision makes the platform more likely to approve safe borrowers and less likely to approve risky ones. The bank consequently loses additional safe lending opportunities, while retaining more of its potential risky loans when those borrowers are also contested. These direct effects need not match the comparison of optimized portfolios, since the bank subsequently adjusts its rate and screening. The adoption-sorting extension in Appendix C shows that the same logic applies when platform users are concentrated in particular parts of the borrower-preference distribution.

The mechanism has counterparts in the evidence on ecosystem lending. Using Mercado Libre data from Argentina, Frost et al. (2019) find that transaction-based credit scores predict loan losses more accurately than conventional credit-bureau ratings. On the other hand, Chen et al. (2023) document reductions in lending and tighter loan-to-value ratios at traditional *non-bank* lenders facing greater BigTech penetration in China, consistent with selective displacement. This is further evidence also on the potential role that scalability of proprietary data by the platform may have on incumbent intermediaries, other than regulated banks. Neither study directly identifies the change in the default frequency of bank-originated loans. Rather, our last proposition characterizes the portfolio-composition

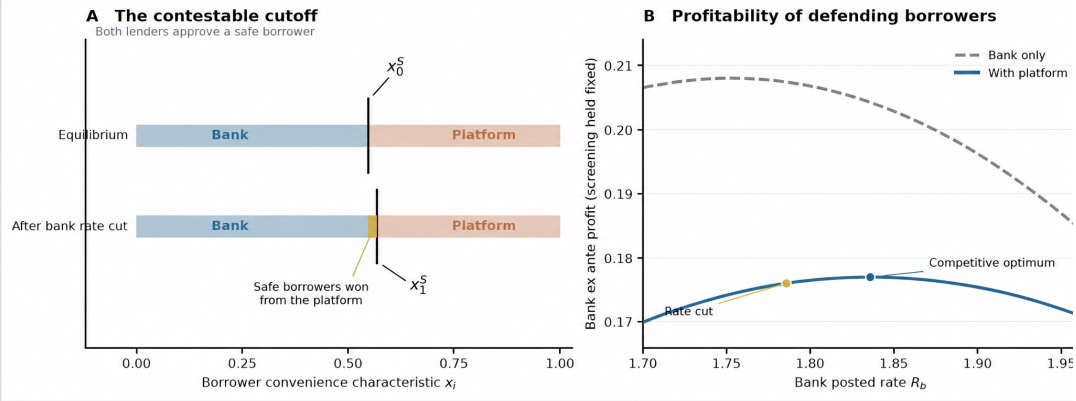


Figure 5: **Market stealing and the bank's common rate.** Panel A shows the safe borrowers won by a bank-rate cut; Panel B compares the bank's profit at different rates, keeping screening fixed. The rate that wins additional borrowers need not maximize the bank's overall lending profit.

effect that would need to be tested in matched incumbent–platform lending data.

5.3 Why does competition not necessarily lower the bank's rate?

A reduction in the bank's rate can attract safe borrowers close to the lender-indifference cutoff $\bar{x}^S$. Figure 5 illustrates that conventional market-stealing incentive. The bank, however, cannot confine a discount to those borrowers: it must offer the lower rate on loans it would have retained without cutting its price.

The distinction is reflected in the bank's profit. In the benchmark competition region, the bank-only profit at given (R_b, I) is reduced by the expected return from safe loans diverted to the entrant:

$$\Pi_b(R_b, I) = \Pi_b^M(R_b, I) - \mu p \gamma \delta \Delta_S m_b^S. \tag{5.3}$$

The bank's pricing condition highlights the trade-off:

$$\frac{\partial \Pi_b^M}{\partial R_b} = \mu p \gamma \delta \theta^S \left[\Delta_S - \frac{1+\rho_p}{2t} m_b^S \right], \tag{5.4}$$

where ρ_p summarizes the platform's rate response. On the right, the first term reflects the value of safe loans transferred to the platform; the second captures how a change in rates

alters the extent of that transfer. The bank weighs these effects against the familiar monopoly trade-off on the left: a higher rate raises repayments on retained loans but discourages participation.

Proposition 2 (Incumbent pricing)**.** *In the interior benchmark competition region, the bank's optimal rate at a given screening intensity exceeds its monopoly-case rate:* $R_b^C(I) > R_b^M(I)$.

The proposition identifies why winning some borrowers is not necessarily a reason to lower the *common* rate. At the bank-only optimum, the bank has already balanced the revenue gained on retained loans against the participation lost from a higher price. Platform entry changes that balance: some of the safe borrowers who would be attracted by a rate cut would ultimately borrow from the platform, whereas the discount applies to bank customers who would have borrowed anyway. Moreover, a rate cut can expand lending to risky applicants whose loans have negative expected margins. Under the benchmark's participation conditions, these forces outweigh the incremental return from defending the cutoff, so the bank prefers a higher conditional rate. Appendix B.11 establishes the comparison from (5.4).

The distinction is between price competition for a marginal borrower and the profitability of repricing the entire loan book. In Broecker (1990), a small undercut at equal rates can transfer a positive mass of jointly approved applicants; here it moves an aggregate cutoff and wins only a small mass. In the framework by Hauswald and Marquez (2006), lenders can bid for an identified borrower, whereas the bank in our model makes a single offer to its applicant population. These differences help explain why fighting for the marginal borrower need not dominate the bank's optimal price.

The higher-rate conclusion is specific to the participation region in which it is established. When bank-only safe demand is saturated, a further rate cut cannot create additional exclusive safe applications but can still win borrowers receiving two offers. The incentive for defensive pricing can therefore be stronger. Appendix B.6 examines the relevant participation boundaries and presents an equilibrium in which the bank charges less than under

monopoly. The purpose of that example is to show that the model can also generate the defensive-price logic emphasized elsewhere in the literature when the participation regime changes.

Observed BigTech lending rates also caution against equating competition with a single ordering of prices. Liu et al. (2022) find that BigTech business loans in China are typically smaller, shorter-lived and charged at higher annualized rates than conventional bank loans, including among borrowers with access to bank credit. These differences are consistent with the importance of relative conveniences or preferences for a loan other than just the pure pricing differences. However, neither identify the incumbent's rate response to platform entry. The testable pricing implication of our model concerns that response, conditional on how many borrowers can be contested and how widely the bank must apply its rate.

5.4 Why is screening less responsive than pricing?

Screening continues to protect the bank from loss-making risky loans. Entry changes the return to that investment because some of the safe borrowers identified by the bank obtain platform credit instead. The screening first-order condition makes the two additional effects explicit:

$$C'(I) = \Lambda(R_b) - \underbrace{\frac{\mu p \delta}{2} \Delta_S m_b^S}_{\text{displaced safe lending}} + \underbrace{\frac{\mu p \gamma \delta \theta^S}{2t} m_b^S \frac{\partial R_p^{\text{br}}}{\partial I}}_{\text{platform repricing}}. \tag{5.5}$$

The bank-only return $\Lambda(R_b)$ rewards more accurate approvals: additional safe projects are financed, and fewer risky projects are mistakenly accepted. Platform competition reduces the first benefit, because some newly identified safe borrowers choose the platform instead. This loss is measured by the second term. The last term works in the opposite direction in the benchmark: better bank screening induces a platform price response that allows the bank to retain more profitable lending. The same information investment therefore improves selection while changing the terms of competition for borrowers.

The equilibrium paths for screening, rates, profits and bank default frequency are dis-

played in Figure 7.

This explains why the screening response can be more muted than the pricing response. As adoption rises initially, the loss of profitable safe loans depresses the return to bank screening. Later, the contested overlap becomes smaller and the platform's pricing adjustment offsets more of that loss. Screening recovers slightly as a result. The curvature of the path in Figure 7 reflects this balance in the benchmark rather than a universal prediction of convex screening; supporting calculations are in Appendix B.12.

5.5 What does platform adoption change in equilibrium?

Platform adoption affects equilibrium through two connected margins: it changes the composition of the borrowers seeking credit from each intermediary, and it changes the allocation of originated loans across lenders. These two objects need not coincide. Borrowers may apply to both lenders, but origination depends on approval, relative loan terms and the platform's informational advantage.

Figure 6 summarizes this mechanism. Panel A shows how the composition of applicant pools changes with platform adoption. Panels B and C then trace how those application patterns translate into realized lending. Safe borrowers are displaced from bank lending more strongly than risky borrowers, while aggregate credit expands as the platform originates loans that would not have been financed by the bank alone.

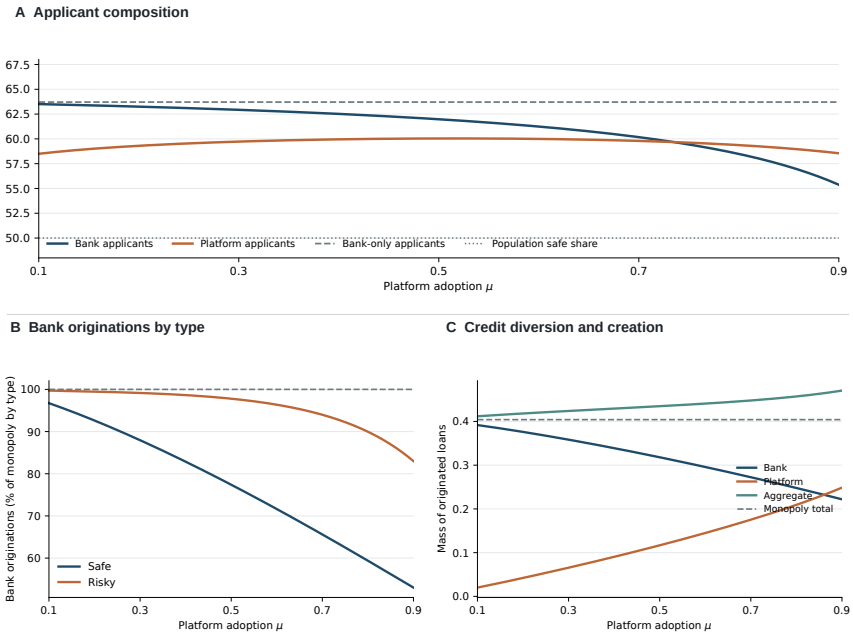


Figure 6: **From applications to originations.** Panel A compares the safe shares of bank and platform applicants. Panel B shows bank originations by true type relative to the optimized bank-only benchmark. Panel C reports bank, platform, and total originated lending.

The bank's rate rises above monopoly while screening responds moderately; its profit declines as the default frequency of retained loans increases (Figure 7). The comparison emphasizes that changes in portfolio quality reflect both the bank's decisions and the reallocation of approved borrowers.

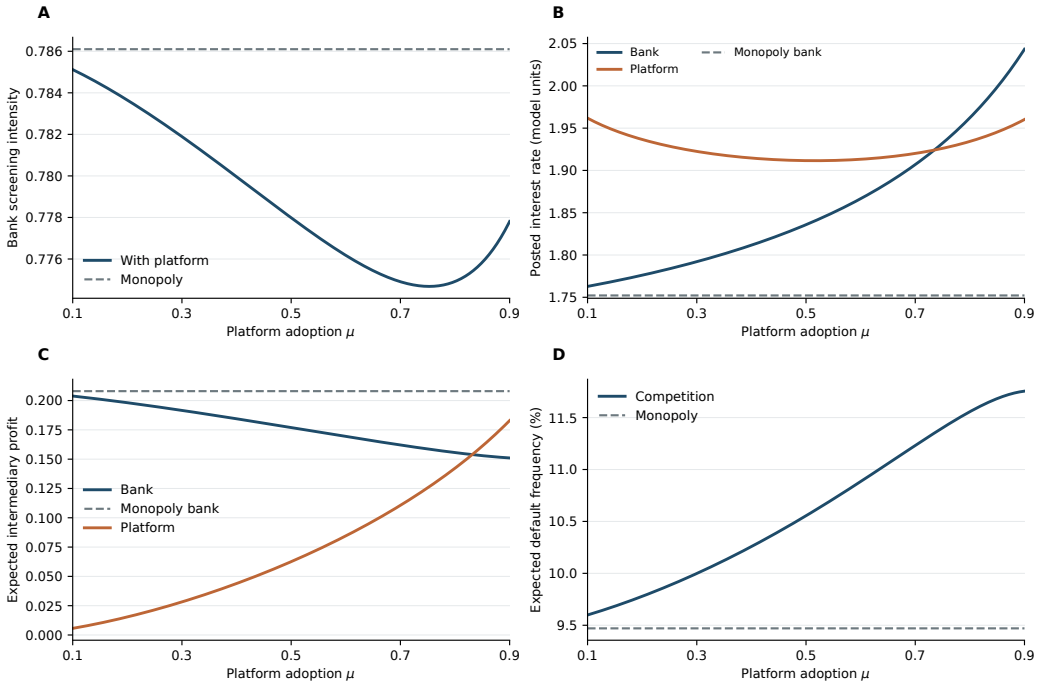


Figure 7: **Competition and the bank-only benchmark.** Screening intensity, posted rates, intermediary profits and the bank's expected default frequency as platform adoption increases; bank-only reference lines and the original calibration are unchanged.

The mechanism suggests a natural empirical distinction between applicants and originated loans. In markets where platform users overlap more with a bank's profitable borrowers, growing platform penetration should generate stronger relative displacement of safer bank loans, especially when the platform's proprietary data are more predictive of repayment. Measuring this channel requires observing both the borrowers seeking credit and those ultimately financed, while accounting for changes in bank rates and screening. The cited studies document important elements of platform information and credit displacement, but do not by themselves establish this bank-portfolio prediction.

Adoption consequently matters through both the borrowers accessible to the platform and the quality of its signal. The possibility than the bank strategically decides to choose interest rates and screening intensity to try to deter platform entry is developed in Appendix B.7 . The nature of competition in our case is showed to not produce high enough incentives for

the bank to try to keep the platform off the market.

6 Comparative Statics

Platform entry changes the composition of credit retained by the bank. We now distinguish the role of the platform's information technology from that of funding costs. The first exercise holds the platform's reach fixed while changing its screening accuracy; the second changes the terms on which both intermediaries can finance loans.

6.1 How does better platform information affect the incumbent's portfolio?

A higher α increases the precision $\delta(\mu;\alpha)$ of the platform's signal at a given adoption level, without changing the population it can reach. Greater precision therefore affects which borrowers receive competing offers rather than the platform's potential market size. In the benchmark coverage region, the additional offers displace safe bank loans, whereas risky bank originations remain unchanged at fixed lender decisions. From the loan masses in (5.1),

$$\left.\frac{\partial \Omega_b}{\partial \alpha}\right|_{R_b,R_p,I} = \frac{(\theta^S-\theta^R)\mu p\gamma\Delta_S L_b^R}{(L_b^S+L_b^R)^2}\,\delta_\alpha > 0, \qquad \delta_\alpha = -\frac{\mu^{1/\alpha}\log\mu}{2\alpha^2} > 0. \tag{6.1}$$

The expression isolates the effect of better information from the bank's subsequent adjustments. The argument extends to markets in which risky borrowers also have overlapping acceptable offers: more accurate platform screening then rejects more risky borrowers, leaving additional risky lending opportunities with the bank.

Proposition 3 (Platform information and bank portfolio selection)**.** *At unchanged lending decisions, greater platform signal precision raises the risky share of bank loans whenever both types are financed and a positive mass of loans is contested. It also weakly lowers the bank's ex ante profit and gross marginal return to screening.*

In the considered lending region, safe-loan margins are positive and the risky-loan margins are negative. The effects arise because the platform becomes more likely to approve safe borrowers and less likely to approve risky ones. For the bank, more precise platform information both reduces the return to identifying a profitable safe applicant and, where risky offers overlap, leaves more mistakenly funded risky applicants available. Appendix D.1 establishes the result for either type of market coverage. The proposition concerns the direct allocation and information-acquisition effects; the equilibrium screening response also incorporates the platform's pricing reaction.

Figure 8 compares $\alpha \in \{1.25, 1.50, 1.75, 2\}$ under the original funding costs. Better platform information is associated with a higher incumbent rate and portfolio default frequency, alongside lower bank profit. The screening response is small by comparison. The bank retains a strong reason to screen to avoid loss-making risky loans, even though platform competition erodes the informational rents generated by identifying safe borrowers. At high adoption, platform repricing and the smaller interval of contestable offers partly offset the direct screening effect, so the equilibrium screening paths need not be ordered monotonically in α.

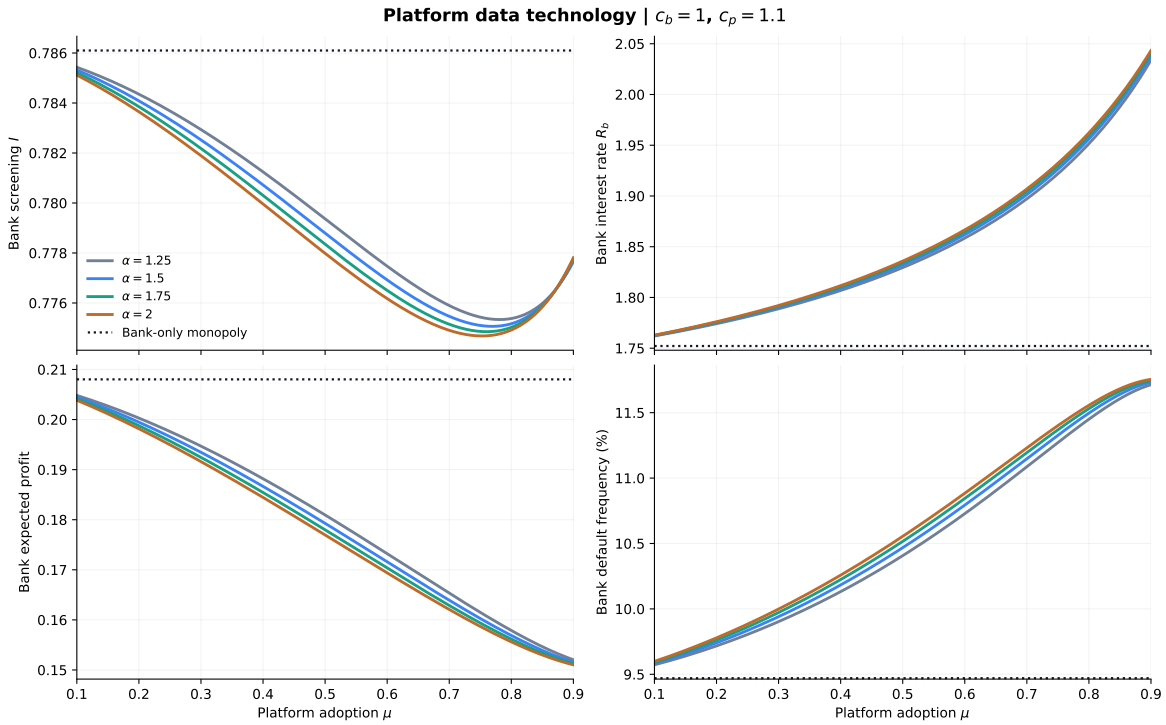


Figure 8: **Platform information and bank outcomes.** Screening intensity, posted bank rate, bank profit, and bank portfolio default frequency for $\alpha \in \{1.25, 1.50, 1.75, 2\}$, with $c_b = 1$ and $c_p = 1.1$. Horizontal reference lines are the same-parameter monopoly equilibrium.

Adoption amplifies the allocation effect because it changes both platform reach and precision. At fixed rates, exposure of safe bank lending to successful platform screening increases with $\mu\delta(\mu;\alpha)$, whose marginal response to adoption is $\delta + \mu\delta_\mu$. Unlike the introduction of a lender with a fixed-quality signal, the expansion of the platform's user base also improves its ability to select borrowers. The numerical equilibrium captures how bank pricing and screening mediate this direct effect.

6.2 When do funding costs change the extent of competition?

Funding costs change expected lending margins and, through rates, the overlap of acceptable offers. We compare the original $c_b = 1$ with $c_b = 1.1$, maintaining $c_p = \psi c_b$ and $\psi = 1.1$. Both lenders' funding costs therefore rise, and each competitive equilibrium is compared

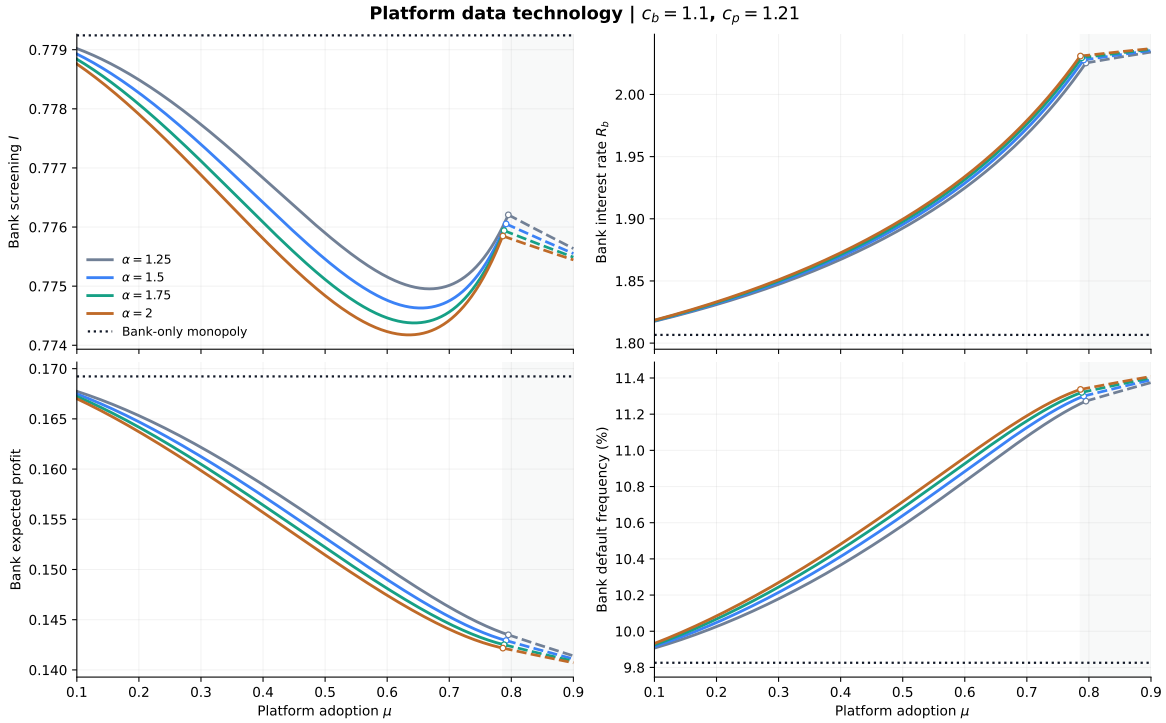


Figure 9: **Platform information under higher funding costs.** The same outcomes for $c_b = 1.1$, $c_p = 1.21$, with $\alpha \in \{1.25, 1.50, 1.75, 2\}$. Horizontal lines use the reoptimized monopoly at $c_b = 1.1$. Dashed portions indicate the binding safe-coverage frontier, not mixed strategies.

with its own same-cost monopoly benchmark.

Within the interior competition region, higher funding costs increase the bank's conditional optimal rate. But the equilibrium effect need not obey a uniform pass-through rule: higher repayments also shrink the group of safe borrowers willing to accept either offer. When $2Y^S - \theta^S(R_b + R_p) = t$, the two acceptable-offer regions cease to overlap. The interior pricing condition then gives way to one-sided conditions at the boundary. Appendix D.2 derives this distinction.

Figure 9 shows that this boundary is reached at high adoption, earlier when the platform's information is more effective. The associated change in the bank's rate and screening responses reflects a change in the set of borrowers the lenders can directly contest. The transition is not, by itself, evidence of a sudden worsening of bank portfolio risk. Nor does the interior pass-through result establish the sign of equilibrium rate changes across the bound-

ary; the two cost configurations may even have very similar bank rates at high adoption.

Taken together, the exercises distinguish information-driven selection from the participation effects of funding costs. More effective platform information increases the pressure on profitable bank lending opportunities, whereas tighter funding conditions change both margins and the extent of competition for those opportunities. These interactions help explain why the incumbent's rates, screening, and retained portfolio risk need not respond proportionately to platform expansion.

7 Efficient credit allocation and welfare

Platform entry may worsen the credit quality of the bank's retained portfolio while expanding credit to previously excluded borrowers. We examine whether this reallocation improves productive efficiency: the expected output of financed projects, net of the real resources used in funding and screening. We focus on productive efficiency in the main text and distinguish it from total welfare below.

7.1 What prevents an efficient allocation of credit?

Consider first the expected value of projects net of funding and screening resources. With $Y^S > c_p > c_b > Y^R$, a planner who observes true borrower types funds all safe projects through the lower-cost bank and rejects all risky projects. Screening is unnecessary. Productive surplus in this first-best allocation is $W^{FB} = p(Y^S - c_b)$, which does not depend on platform adoption.

In the decentralized economy, net productive surplus is

$$W^P = \sum_{j \in \{S,R\}} \left[L_b^j (Y^j - c_b) + L_p^j (Y^j - c_p) \right] - C(I). \tag{7.1}$$

Expected project output Y^j already reflects default risk, so unsuccessful repayment is not

deducted again. Funding costs and screening expenditure are real resource costs; screening is paid once, rather than for each applicant. The distance from the first-best (W^{FB}) can be decomposed as follows:

$$\begin{aligned} W^{FB} - W^{P} = (Y^{S} - c_b)\left[p - L_b^S - L_p^S\right] + (c_p - c_b)L_p^S \\ + (c_b - Y^R)L_b^R + (c_p - Y^R)L_p^R + C(I). \end{aligned} \tag{7.2}$$

The first term measures productive safe projects left unfunded. The second is the additional funding cost of safe loans originated by the platform rather than the bank. The remaining terms measure the social cost of financing risky projects and the resources devoted to screening. All are nonnegative under the maintained assumptions. Imperfect information produces both mistaken risky lending and the rejection of safe projects; borrower participation and common-rate pricing further shape the extent of unfinanced safe credit. Equation (7.2) therefore separates the fundamental screening friction from the allocation and funding-cost consequences of competition.

7.2 Does platform entry improve credit allocation?

The first-best is independent of μ, but the privately informed equilibrium is not. Platform lending can finance safe projects that the bank would not have originated, while also replacing bank loans with more expensive platform credit. These effects are accompanied by changes in the bank's screening and rate. There is consequently no presumption that productive surplus must rise with platform adoption.

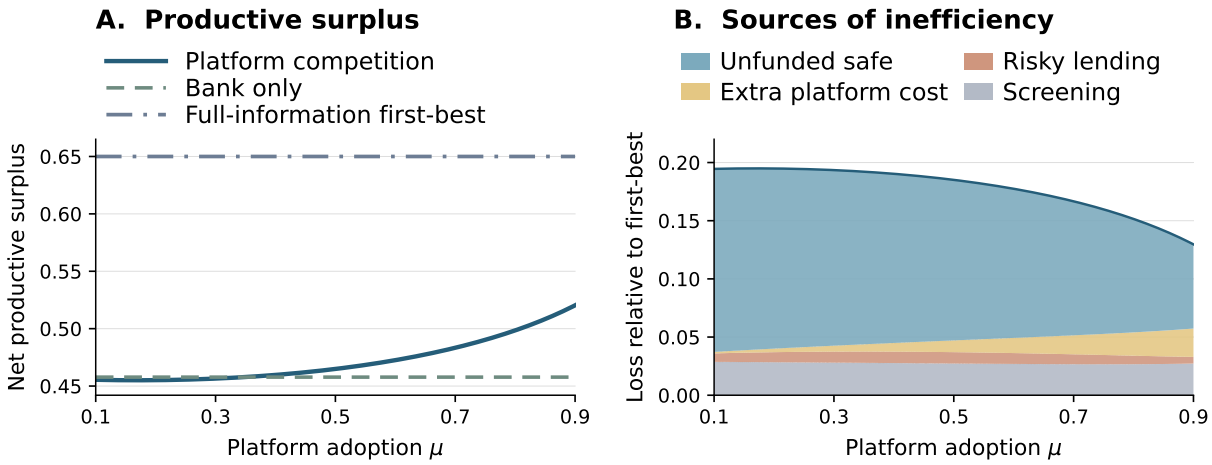


Figure 10: **Productive efficiency and its sources.** Panel A compares equilibrium net productive surplus with the bank-only equilibrium and the full-information first-best. Panel B decomposes the difference from the first-best into unfinanced safe projects, the platform's additional funding cost, mistaken financing of risky projects, and screening expenditure. All curves use the same benchmark equilibrium as the preceding sections.

Figure 10 shows that the largest efficiency loss comes from safe projects remaining unfunded. As adoption expands, the platform increasingly closes this gap, even as it diverts some safe lending toward a more costly intermediary. Under the benchmark, productive surplus is slightly below the bank-only level at low adoption, reaches its minimum near $\mu = 0.17$, and overtakes the bank-only outcome around $\mu = 0.35$. Thereafter it increases over the computed range. The improvement is driven chiefly by additional safe credit, rather than by a large adjustment in bank screening. Thus, the higher expected default frequency of the bank's retained portfolio need not indicate a less efficient allocation of credit in the economy as a whole.

These comparisons concern productive efficiency, not total welfare. If borrower convenience is treated as a genuine social disutility, total welfare is $W = W^P - \mathcal{D}$. The aggregate non-pecuniary cost $\mathcal{D}$ and the corresponding welfare accounting are derived in Appendix E.

8 Conclusion

Digital platforms are pervasive in many markets and through its ecosystem acquire information that could be exploited for entering successfully into new markets. Here we have considered how that information could be useful for entering into credit provision. Unlike incumbent banks, which acquire screening precision through costly investment, a platform's adoption base determines both the borrowers it can reach and the accuracy of its signal. We show how these two consequences of adoption jointly shape credit-market competition and the risk composition of the incumbent's loan portfolio.

The main effect of platform entry is not necessarily a change in the volume of risky bank lending. Rather, the bank's portfolio deteriorates when the entrant disproportionately displaces safe lending opportunities. This result survives even when both borrower types have access to competing acceptable offers. The incumbent's pricing and screening responses need not reverse the effect. In particular, competition for borrowers who hold relative convenience costs for both lenders does not necessarily lower the bank's common rate, since a reduction would also reprice loans the bank would otherwise retain.

The available evidence motivates these mechanisms but does not yet establish their combined implications. Using Mercado Libre data from Argentina, Frost et al. (2019) document the predictive value of information generated by e-commerce activity. Liu et al. (2022) show that BigTech business credit in China differs from bank lending in size, maturity, and interest rates, while Chen et al. (2023) find lending contractions and tighter terms among competing traditional nonbank lenders. Our framework connects these observations through the selection of borrowers across lenders. A direct empirical test would examine how platform penetration and information quality affect the composition of bank applicants and originations, accounting for incumbent pricing and screening responses.

Platform entry can also finance safe projects that would otherwise remain unfunded, improving productive credit allocation despite deterioration in the bank's retained portfolio. Thus, the consequences of BigTech lending cannot be inferred from credit growth, interest

rates, or incumbent default frequencies in isolation. The framework takes ecosystem adoption as given; how adoption and information sharing are determined remains an important question for further research.

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A Bank monopoly: information, ex ante profits, and optimality

A.1 Applicant selection and posterior beliefs

The bank announces a common gross repayment R_b before observing individual applications or screening outcomes. For $j \in \{S, R\}$, let $D_b^j(R_b)$ be the fraction of type-j borrowers willing to apply for a bank loan. With the borrower utility specified in the main text, the exact application probability is

$$D_b^j(R_b) = \left[\frac{Y^j - \theta^j R_b}{t}\right]_0^1, \qquad [z]_0^1 \equiv \min\{1, \max\{0, z\}\}. \tag{A.1}$$

The associated mass of applicants and their safe-type prior are

$$\mathcal{A}_b = pD_b^S + (1-p)D_b^R, \qquad p_A = \Pr(S \mid A_b) = \frac{pD_b^S}{\mathcal{A}_b}, \tag{A.2}$$

where A_b denotes application to the bank and the conditional probability requires $\mathcal{A}_b > 0$. Thus, the composition of applications is potentially different from the population prior p, even before the bank receives any information.

For completeness, define $h_s = \Pr(m = s \mid A_b)$ and $h_r = \Pr(m = r \mid A_b)$. Under $\gamma = \gamma(I) = (1+I)/2$,

$$h_s = p_A\gamma + (1-p_A)(1-\gamma), \qquad h_r = p_A(1-\gamma) + (1-p_A)\gamma. \tag{A.3}$$

The probabilities that an applicant is truly safe after a favorable signal and truly risky after an unfavorable signal are, respectively,

$$\beta^S = \Pr(S \mid A_b, m = s) = \frac{p_A\gamma}{h_s}, \qquad \beta^R = \Pr(R \mid A_b, m = r) = \frac{(1-p_A)\gamma}{h_r}. \tag{A.4}$$

At $I = 0$ these become $\beta^S = p_A$ and $\beta^R = 1 - p_A$. These beliefs describe the information contained in the screening outcomes; *they are not additional weights in ex ante profit.* The latter is computed below directly from the unconditional joint masses of applications, approvals, and ultimately originated loans.

A.2 Loan origination and expected profits

Write $p_S = p$ and $p_R = 1 - p$, and define the expected margin on a loan to a *true* type-j borrower as

$$m_b^j(R_b) \equiv \theta^j R_b - c_b.$$

A type-j borrower applies with probability D_b^j and, conditional on applying, is approved after a favorable signal with probability $\Pr(m = s \mid j)$. Under the approval policy $M_b = \{s\}$, these probabilities are γ for safe borrowers and $1 - \gamma$ for risky borrowers. Hence the ex ante originated masses are exactly those of equation (4.1):

$$L_0^S = \underbrace{p}_{\text{safe share}} \underbrace{D_b^S}_{\text{application}} \underbrace{\gamma}_{\text{approval}}, \qquad L_0^R = \underbrace{(1-p)}_{\text{risky share}} \underbrace{D_b^R}_{\text{application}} \underbrace{(1-\gamma)}_{\text{approval}}. \tag{A.5}$$

Expected profit is therefore equation (4.2):

$$\Pi_b^M(R_b, I; \{s\}) = L_0^S m_b^S + L_0^R m_b^R - C(I), \qquad C(I) = \frac{\kappa I^2}{1 - I}. \tag{A.6}$$

No loan-specific funding cost is incurred by rejected applicants, and the information-acquisition cost is paid once, not per application or loan. In particular, equation (A.6) does not multiply L_0^S or L_0^R by the posterior probabilities in (A.4). The unconditional originated-loan masses in (A.5) are the sole weights in the profit expression; posterior beliefs describe signal informativeness.

A.3 Proof of Lemma 1

With positive demand of both types, the risky-to-safe odds among applicants equal $(1-p)D_b^R/(pD_b^S)$. Under favorable-signal approval the corresponding odds among originated loans multiply this expression by $(1-\gamma)/\gamma = (1-I)/(1+I)$. For $0 < I < 1$, the latter factor is strictly below one, establishing the first comparison. Holding the rate fixed,

$$\frac{\partial}{\partial I} \log \frac{L_0^R}{L_0^S} = -\frac{2}{1-I^2} < 0.$$

Since $\theta^S > \theta^R$, equation (4.6) then implies that the risky share and expected default frequency decrease strictly with screening. This partial effect does not sign the outcome when the posted rate or approval policy is reoptimized.

A.4 Alternative approval policies

Let $e_s, e_r \in \{0,1\}$ indicate whether the bank approves a borrower after signals s and r, respectively. The conditional offer probabilities are

$$q_b^S = e_s\gamma + e_r(1-\gamma), \qquad q_b^R = e_s(1-\gamma) + e_r\gamma. \tag{A.7}$$

For any R_b and I, the ex ante objective for the corresponding policy is

$$\Pi_b^M(R_b, I; e_s, e_r) = pD_b^S q_b^S m_b^S + (1-p)D_b^R q_b^R m_b^R - C(I). \tag{A.8}$$

This formula covers favorable-only lending $(e_s, e_r) = (1,0)$, unfavorable-only lending $(0,1)$, pooling $(1,1)$, and no lending $(0,0)$. Under pooling, information does not affect approval probabilities and $I = 0$ is optimal; under no lending, $I = 0$ and profits are zero.

Assumption 2 implies that any risky applicant willing to accept the bank contract has $m_b^R < 0$. When $m_b^S \geq 0$, the profit difference at a common (R_b, I) between approving only

favorable signals and approving only unfavorable signals is

$$\Pi_b^M(R_b, I; 1, 0) - \Pi_b^M(R_b, I; 0, 1) = (2\gamma - 1)\left[pD_b^S m_b^S - (1-p)D_b^R m_b^R\right] \geq 0. \tag{A.9}$$

When $m_b^S < 0$, both true-type margins are nonpositive for borrowers who accept credit and no lending weakly dominates unfavorable-only lending. Thus, under the maintained primitives, unfavorable-only lending cannot strictly dominate both favorable-only lending and no lending. Nevertheless, the monopoly optimum is obtained only after optimizing and comparing the payoffs of the remaining policies over all admissible prices and screening intensities, including participation boundaries.

A.5 Conditional optimality and global solution

For favorable-signal lending write

$$a_S(I) \equiv p\gamma(I), \qquad a_R(I) \equiv (1-p)[1-\gamma(I)].$$

Then $\Pi_b^M = \sum_j a_j(I) D_b^j m_b^j - C(I)$. In any region where $0 < D_b^S, D_b^R < 1$, differentiation at fixed I gives the interest-rate condition stated in equation (4.3):

$$0 = \frac{\partial \Pi_b^M}{\partial R_b} = \sum_{j \in \{S,R\}} a_j(I) \left[\theta^j D_b^j - \frac{\theta^j}{t} m_b^j\right] \tag{A.10}$$

$$= \frac{1}{t} \sum_{j \in \{S,R\}} a_j(I) \theta^j \left(Y^j + c_b - 2\theta^j R_b\right). \tag{A.11}$$

Here the first term in square brackets is the marginal revenue effect on existing loans and the second captures the endogenous reduction in applications. The conditional optimal price within this interior cell is

$$R_b^M(I) = \frac{a_S(I)\theta^S(Y^S + c_b) + a_R(I)\theta^R(Y^R + c_b)}{2\left[a_S(I)(\theta^S)^2 + a_R(I)(\theta^R)^2\right]}. \tag{A.12}$$

At a fixed R_b, since $\gamma'(I) = 1/2$, the screening condition in equation (4.4) is

$$\frac{\partial \Pi_b^M}{\partial I} = 0 \quad \Longleftrightarrow \quad \underbrace{\frac{1}{2}\left[pD_b^S m_b^S - (1-p)D_b^R m_b^R\right]}_{\Lambda(R_b)} = \underbrace{\kappa\left[\frac{1}{(1-I)^2} - 1\right]}_{C'(I)}. \tag{A.13}$$

The screening benefit $\Lambda(R_b)$ includes the gains from identifying safe borrowers and avoiding negative-margin risky borrowers. $C''(I) = 2\kappa/(1-I)^3 > 0$, so for each fixed rate the unique favorable-policy screening choice is

$$I^M(R_b) = \begin{cases} 1 - \sqrt{\dfrac{\kappa}{\kappa + \Lambda(R_b)}}, & \Lambda(R_b) > 0, \\ 0, & \Lambda(R_b) \leq 0. \end{cases} \tag{A.14}$$

An equivalent one-dimensional characterization profiles screening out:

$$\max_{I \in [0,1)} \Pi_b^M(R_b, I; \{s\}) = \frac{K_S(R_b) + K_R(R_b)}{2} + \left[\sqrt{\kappa + \max\{\Lambda(R_b), 0\}} - \sqrt{\kappa}\right]^2, \tag{A.15}$$

where $K_S = pD_b^S m_b^S$ and $K_R = (1-p)D_b^R m_b^R$. This expression continues to hold when either demand is zero or one. The full monopoly optimum compares its global maximum over R_b with pooling and nonlending. The conditions (A.10) and (A.13) apply only at a smooth interior optimum.

At participation boundaries, which occur at $R_b = (Y^j - t)/\theta^j$ and $R_b = Y^j/\theta^j$, D_b^j remains continuous but has a kink. On a differentiable segment its rate derivative is $-(\theta^j/t)$ if $0 < D_b^j < 1$ and zero when demand is locally constant. The exact profit derivative is then

$$\frac{\partial \Pi_b^M}{\partial R_b} = \sum_j a_j \left[\theta^j D_b^j + m_b^j \frac{\partial D_b^j}{\partial R_b}\right].$$

At a candidate interior kink R_0, the necessary one-sided condition is $\Pi'_{b,-}(R_0) \geq 0 \geq \Pi'_{b,+}(R_0)$, with one-sided derivatives of the optimized, screened payoff when screening is profiled out. A global maximum must still be compared across the smooth cells and ap-

proval policies.

A.6 Rate, screening, and portfolio-risk implications

Within the interior favorable-policy cell, the conditional rate responds positively to bank funding cost:

$$\left.\frac{\partial R_b^M(I)}{\partial c_b}\right|_I = \frac{a_S(I)\theta^S + a_R(I)\theta^R}{2\left[a_S(I)(\theta^S)^2 + a_R(I)(\theta^R)^2\right]} > 0. \tag{A.16}$$

For the conditional response to screening, define $w_j = a_j(I)(\theta^j)^2$ and $r_j = (Y^j + c_b)/(2\theta^j)$. Direct differentiation of (A.12) yields

$$\frac{dR_b^M(I)}{dI} = \frac{p(1-p)(\theta^S)^2(\theta^R)^2}{2(w_S + w_R)^2}(r_S - r_R). \tag{A.17}$$

This derivative is negative when $r_S < r_R$, as in the benchmark; it is not necessarily negative for all primitives.

For favorable-signal bank lending, the expected default frequency is

$$\Omega_b^M = \frac{L_0^S(1-\theta^S) + L_0^R(1-\theta^R)}{L_0^S + L_0^R} = (1-\theta^S) + (\theta^S - \theta^R)\frac{L_0^R}{L_0^S + L_0^R}. \tag{A.18}$$

Moreover, conditional on positive loan masses,

$$\frac{L_0^R}{L_0^S} = \frac{1-p}{p}\frac{1-I}{1+I}\frac{D_b^R}{D_b^S}.$$

Thus portfolio composition depends separately on population odds, screening accuracy, and borrower selection into applications. With interior demand and $R_b = R_b^M(I)$,

$$\frac{d}{dI}\log\frac{L_0^R}{L_0^S} = -\frac{2}{1-I^2} + \left[\frac{\theta^S}{Y^S - \theta^S R_b} - \frac{\theta^R}{Y^R - \theta^R R_b}\right]\frac{dR_b^M(I)}{dI}. \tag{A.19}$$

If $\theta^S Y^R \geq \theta^R Y^S$ and $dR_b^M/dI \leq 0$, this derivative is negative. The qualification matters: for arbitrary primitives, the endogenous rate response need not reinforce the direct screening

effect.

Finally, at a strict smooth interior optimum the screening-cost comparative static follows from the implicit condition $\Lambda(R_b^M(I)) = C'(I)$. With $F(I,\kappa) = \Lambda(R_b^M(I)) - \kappa[(1-I)^{-2}-1]$ and $F_I < 0$,

$$\frac{dI^M}{d\kappa} = \frac{(1-I^M)^{-2}-1}{F_I} < 0.$$

The implied rate and risk responses inherit the stated conditional sign restrictions. None of these derivatives establishes global monotonicity across changes in participation or approval policies.

B Platform competition: ex ante loan masses and profit accounting

B.1 Joint events and conditional demands

Retain $p_S = p$, $p_R = 1-p$ and the conditional participation probabilities D_b^j, D_p^j defined in Section 3. Because lenders cannot condition their rates on private convenience characteristics, an eligible type-j borrower can apply to both lenders. Let $\widetilde{D}_\ell^j$ denote the share choosing lender ℓ when both lenders extend an acceptable offer. For an arbitrary pair of posted rates these quantities are computed from

$$\bar{x}^j = \frac{1}{2} + \frac{\theta^j(R_p - R_b)}{2t}, \qquad \widetilde{D}_b^j = \min\{D_b^j, [\bar{x}^j]_0^1\}, \qquad \widetilde{D}_p^j = \min\{D_p^j, [1-\bar{x}^j]_0^1\}.$$

These formulas incorporate the outside option and permit a gap in acceptable offers. They reduce to $\bar{x}^j$ and $1-\bar{x}^j$ only in an interior covered market. In particular, when the acceptable offer sets are disjoint, $\widetilde{D}_b^j = D_b^j$ and $\widetilde{D}_p^j = D_p^j$.

Let q_b^j be the probability of a bank offer conditional on a potential type-j application and

let q_p^j be the corresponding platform probability. Under arbitrary signal-approval policies,

$$q_b^S = e_{b,s}\gamma + e_{b,r}(1-\gamma), \quad q_b^R = e_{b,s}(1-\gamma) + e_{b,r}\gamma,$$

$$q_p^S = e_{p,s}\delta + e_{p,r}(1-\delta), \quad q_p^R = e_{p,s}(1-\delta) + e_{p,r}\delta,$$

where $e_{\ell,k} \in \{0,1\}$. The platform can lend only to adopters, of mass μ, while the bank can lend to all borrowers. Both signals are independent conditional on the true type, and adoption is independent of true type and convenience.

B.2 Originations and expected profits

Partition type-j bank lending by whether a competing platform offer is made. Ex ante origination masses are

$$L_b^j = \underbrace{p_j q_b^j (1-\mu q_p^j) D_b^j}_{\text{no platform offer}} + \underbrace{p_j \mu q_b^j q_p^j \widetilde{D}_b^j}_{\text{both offers; bank chosen}}, \tag{B.1}$$

$$L_p^j = \underbrace{\mu p_j q_p^j (1-q_b^j) D_p^j}_{\text{no bank offer}} + \underbrace{\mu p_j q_p^j q_b^j \widetilde{D}_p^j}_{\text{both offers; platform chosen}}. \tag{B.2}$$

The two terms for a given intermediary are mutually exclusive; loans are neither omitted nor counted twice. Integrating over the unobserved convenience characteristic is already contained in the D and $\widetilde{D}$ terms.

Since L_ℓ^j is the *unconditional originated mass of true type* j, expected profits are simply

$$\Pi_b = \sum_{j\in\{S,R\}} L_b^j(\theta^j R_b - c_b) - C(I), \qquad \Pi_p = \sum_{j\in\{S,R\}} L_p^j(\theta^j R_p - c_p). \tag{B.3}$$

No applicant posterior enters this expression as a separate weight. Posterior beliefs govern the informational interpretation of the signal; the joint offer and choice probabilities determine the numbers of loans actually originated.

Under the favorable-only policies $M_b = M_p = \{s\}$, one has $(q_b^S, q_b^R) = (\gamma, 1-\gamma)$ and $(q_p^S, q_p^R) = (\delta, 1-\delta)$. Substitution into (B.1) reproduces the four loan masses in Section 5.1. In the benchmark safe-covered/risky-uncovered region, $\widetilde{D}_b^R = D_b^R$ and $\widetilde{D}_p^R = D_p^R$, whence

$$L_b^S = p\gamma\big[D_b^S - \mu\delta(D_b^S - \widetilde{D}_b^S)\big], \qquad L_b^R = (1-p)(1-\gamma)D_b^R, \tag{B.4}$$

$$L_p^S = \mu p\delta\big[(1-\gamma)D_p^S + \gamma\widetilde{D}_p^S\big], \qquad L_p^R = \mu(1-p)(1-\delta)D_p^R.$$

Defining $\Delta_S \equiv D_b^S - \widetilde{D}_b^S$ and using the monopoly profit function evaluated at the *same* bank rate and screening investment yields the exact identity

$$\Pi_b(R_b, I; R_p) = \Pi_b^M(R_b, I) - \mu p\gamma(I)\delta\Delta_S\,(\theta^S R_b - c_b), \tag{B.5}$$

which reproduces equation (5.2) after substituting the platform's best-response rate $R_p^{br}(R_b, I)$. This is not a comparison of reoptimized monopoly and competitive profits.

B.3 Conditional signal probabilities and lender choice

For clarity, the probability that a bank applicant is safe conditional on receiving a favorable bank signal remains β^S from Appendix A.1: application to the bank depends on its posted rate, not on the realization of the unobserved platform signal. The platform's applicant prior is

$$p_{A,p} = \frac{pD_p^S}{pD_p^S + (1-p)D_p^R},$$

when its applicant mass is positive. Its corresponding favorable-signal safe posterior is

$$\beta_p^S = \frac{p_{A,p}\delta}{p_{A,p}\delta + (1-p_{A,p})(1-\delta)}.$$

These posterior probabilities generally differ from the true-type composition of originated loans, because the latter also reflects competition after both signals have been realized:

$$\Pr(S \mid \text{bank loan}) = \frac{L_b^S}{L_b^S + L_b^R}, \qquad \Pr(S \mid \text{platform loan}) = \frac{L_p^S}{L_p^S + L_p^R}.$$

The distinction between applicants, approved borrowers and originated loans is therefore essential for both profits and bank portfolio risk. Global price optimization must use the piecewise conditional demands above; the interior optimality conditions in Section 5 do not apply at participation boundaries.

B.4 Platform pricing and best responses

The ex ante loan masses and profit functions are defined in equations (B.1) and (B.3); these definitions apply to every approval policy and pricing region.

The platform's exact value after observing the bank commitment is

$$V_p(R_b, I, M_b; \mu) = \max\left\{0, \sup_{R_p,\, M_p \neq \varnothing} \Pi_p(R_p, M_p; R_b, I, M_b)\right\}.$$

The first alternative is no lending. The platform's global best response must compare all admissible offer policies $M_p \in \{\{s\}, \{r\}, \{s, r\}\}$ and all applicable pricing cells.

In the safe-covered/risky-uncovered region with $M_b = M_p = \{s\}$, write $\gamma = \gamma(I)$ and $\delta = \delta(\mu)$. The relevant platform volumes simplify to

$$\frac{L_p^S}{\mu p \delta} = (1-\gamma) D_p^S + \gamma(1 - \bar{x}^S), \qquad \frac{L_p^R}{\mu(1-p)(1-\delta)} = D_p^R.$$

Set $w_S = p\delta$, $w_R = (1-p)(1-\delta)$, and define

$$u_S = (1-\gamma)\frac{Y^S}{t} + \frac{\gamma}{2} + \frac{\gamma \theta^S R_b}{2t}, \quad v_S = \frac{\theta^S}{t}\left(1 - \frac{\gamma}{2}\right),$$
$$u_R = \frac{Y^R}{t}, \qquad\qquad\qquad\quad v_R = \frac{\theta^R}{t}.$$

Then $L_p^j = \mu w_j(u_j - v_j R_p)$ and $\Pi_p = \mu \sum_j w_j(\theta^j R_p - c_p)(u_j - v_j R_p)$. Differentiating gives

$$0 = \frac{1}{\mu}\frac{\partial \Pi_p}{\partial R_p} = \sum_{j \in \{S,R\}} w_j \left[\theta^j u_j + c_p v_j - 2\theta^j v_j R_p\right]. \tag{B.6}$$

Its second derivative is $-2\mu \sum_j w_j \theta^j v_j < 0$ for $\mu > 0$. The conditional interior best response is therefore

$$R_p^{\mathrm{br}}(R_b, I) = \frac{\sum_j w_j(\theta^j u_j + c_p v_j)}{2\sum_j w_j \theta^j v_j}, \qquad \rho_p \equiv \frac{\partial R_p^{\mathrm{br}}}{\partial R_b} = \frac{w_S \gamma (\theta^S)^2}{4t \sum_j w_j \theta^j v_j} > 0.$$

Here μ affects the conditional price through $\delta(\mu)$ and the bank's response, rather than its multiplicative effect on platform size. The conditional optimum is valid only if the resulting prices satisfy the assumed coverage inequalities and outperform alternative platform approval policies and pricing cells. In particular, the platform's rate cannot be inferred from (B.6) when its best response is at a kink or is nonentry.

B.5 Competition: the bank's problem, first-order conditions, and equilibrium

With favorable-signal offers, define $a_S = p\gamma$, $a_R = (1-p)(1-\gamma)$ and $\Delta_S \equiv D_b^S - \widetilde{D}_b^S$. In the benchmark region $L_b^R = a_R D_b^R$, whereas $L_b^S = a_S(D_b^S - \mu\delta\Delta_S)$. The bank's ex ante return, evaluated along the platform's response, is exactly

$$\Pi_b^C(R_b, I) = \Pi_b^M(R_b, I) - \mu a_S \delta \Delta_S m_b^S. \tag{B.7}$$

No assumption about the platform's signal being superior to the bank's is needed for this identity. It expresses the profit from the bank's lost *safe* originations at the same bank choices; it does not compare the two optimized equilibria.

Let $F_M = \partial \Pi_b^M / \partial R_b$ hold screening fixed. Within the smooth benchmark cell, $\Delta_S =$

$(Y^S - t/2 - \theta^S(R_b + R_p)/2)/t$ and, along the platform's price response,

$$\frac{d\Delta_S}{dR_b} = -\frac{\theta^S}{2t}(1 + \rho_p).$$

Thus the bank's conditional price first-order condition is

$$F_M(R_b, I) = \mu a_S \delta \theta^S \left[\Delta_S - \frac{1 + \rho_p}{2t} m_b^S\right]. \tag{B.8}$$

The two terms on the right have distinct economic meanings. The first captures the infra-marginal profitable safe business lost to the platform; the second captures how changing the bank rate and inducing platform repricing alter the amount of that displacement. The presence of direct market stealing therefore does not, by itself, imply a lower optimal common bank rate.

Similarly, $\gamma'(I) = 1/2$ and $d\Delta_S/dI = -(\theta^S/2t)\partial R_p^{\text{br}}/\partial I$. Differentiating (B.7) gives

$$C'(I) = \Lambda(R_b) - \frac{\mu p \delta}{2}\Delta_S m_b^S + \frac{\mu p \gamma \delta \theta^S}{2t} m_b^S \frac{\partial R_p^{\text{br}}}{\partial I}. \tag{B.9}$$

The last term is a strategic response term; it would be absent if the platform set its rate before the bank chose screening. For completeness, if $N(I) = \sum_j w_j(\theta^j u_j + c_p v_j)$ and $H(I) = \sum_j w_j \theta^j v_j$, the response is $\partial_I R_p^{\text{br}} = (N_I - 2R_p^{\text{br}} H_I)/(2H)$, which need not have a uniform sign away from the benchmark.

For each bank choice (R_b, I, M_b), let $\text{BR}_p(R_b, I, M_b)$ be the argmax correspondence of the platform's objective in (B.3), including nonentry. A pure-strategy Stackelberg equilibrium consists of a bank commitment (R_b^*, I^*, M_b^*) and a specified platform best-response selection $\sigma_p \in \text{BR}_p$ such that (R_b^*, I^*, M_b^*) maximizes the bank's induced ex ante profit, with applications, signals and contract choices implemented as specified in the model. If the follower has multiple exact maximizers, a selection or tie-breaking convention is necessary to make the leader's problem single-valued; the numerical benchmark does not rely on an observed strict

follower tie along its interior equilibrium path. Equations (B.6)–(B.9) characterize an *interior candidate* for given approval policies. They are not a substitute for global comparison with boundaries, other approval sets and nonentry.

B.6 Non-interior rates, coverage changes, and one-sided optimality

For any real z, let $[z]_0^1 \equiv \max\{0, \min\{1, z\}\}$. Using the participation and preference thresholds defined in the main text, the correct demands at arbitrary posted rates are

$$D_b^j = [\bar{x}_b^j]_0^1, \quad D_p^j = [1 - \bar{x}_p^j]_0^1, \quad \widetilde{D}_b^j = [\min\{\bar{x}_b^j, \bar{x}^j\}]_0^1, \quad \widetilde{D}_p^j = [1 - \max\{\bar{x}_p^j, \bar{x}^j\}]_0^1. \tag{B.10}$$

The region in which competing offers are both acceptable is determined by $2Y^j - \theta^j(R_b + R_p) \geq t$; the strict reverse inequality gives an uncovered interval. Other boundaries arise when a lender's participation threshold reaches 0 or 1, or when $\bar{x}^j$ reaches the endpoint of the borrower-characteristic interval. Hence a lender's optimal interest rate may lie at a participation kink, at the transition between covered and uncovered demand, or at an admissible endpoint.

For example, when no truncation at 0 or 1 is active, the derivatives of demand for a borrower receiving both offers depend on the constraint that limits the lender's allocation:

binding condition	$\partial \widetilde{D}_b^j / \partial R_b$	$\partial \widetilde{D}_b^j / \partial R_p$
$\bar{x}_b^j < \bar{x}^j$ (bank participation)	$-\theta^j/t$	0
$\bar{x}^j < \bar{x}_b^j$ (relative preference)	$-\theta^j/(2t)$	$\theta^j/(2t)$

binding condition	$\partial \widetilde{D}_p^j / \partial R_b$	$\partial \widetilde{D}_p^j / \partial R_p$
$\bar{x}_p^j > \bar{x}^j$ (platform participation)	0	$-\theta^j/t$
$\bar{x}^j > \bar{x}_p^j$ (relative preference)	$\theta^j/(2t)$	$-\theta^j/(2t)$

All derivatives vanish when the relevant allocation is locally clamped at zero or one. Consequently, the standard price-competition margin applies only when relative preference binds, whereas the margin of new credit participation applies when the lender's own acceptance threshold binds. At equality of the thresholds there is no single classical derivative, and the pertinent one-sided conditions must be used.

On each smooth region, conditional demands are affine functions of the posted rates, so platform profit at a fixed bank commitment and approval policy is quadratic in R_p. The exact follower optimization can therefore compare (i) stationary points inside each smooth pricing cell, (ii) cell endpoints, and (iii) all signal-offer policies. The incumbent then compares its induced payoff across the follower's pricing regions and its own approval policies. At a kink, the correct local condition is a nonnegative left derivative and nonpositive right derivative (with modifications at domain endpoints), not the interior first-order equality. A crossing between follower best responses can also make the leader's induced payoff nonsmooth; the bank's global comparison must retain those candidate points.

One implication is that the higher-rate result in the benchmark interior cell does not rule out lower bank rates in other cells. For a concrete admissible example, retain the benchmark $p = 0.5$, $(\theta^S, \theta^R) = (0.95, 0.25)$, $Y^R = 0.80$, $c_b = 1$, $\kappa = 0.01$, and $\alpha = 2$, and set $Y^S = 2.25$, $t = 0.50$, and $c_p = 1.16$. Both lenders continue to face negative-NPV truly risky projects, $Y^R < c_b < c_p$. At the competitive solution, both credit types have overlapping acceptable contracts when both offers arrive; thus the original risky-uncovered formula does not carry over. In the associated *bank-only* monopoly, the optimal rate is the safe full-participation boundary $(Y^S - t)/\theta^S = 1.8421$, where its smooth interior pricing equation is inapplicable. At $\mu = 0.90$, the competitive optimum has $R_b = 1.8186 < R_b^M$, $I = 0.7814$, and favorable-signal offers by both lenders. The competitive bank default frequency is approximately 13.86%, compared with 9.56% under the same-parameter monopoly. Thus the interest-rate sign can reverse without altering the assumption $Y^R < c_\ell$ or the central portfolio-selection mechanism.

The transition can be assessed using the one-sided marginal-profit conditions at the safe participation boundary; the corresponding illustrative figure is retained in the earlier numerical audit but is not required for the analytical result.

B.7 Entry blockade, deterrence, and accommodation

There are three distinct notions. Entry is *blockaded* at a bank-only monopoly commitment if the entrant's maximal operating profit is nonpositive without any defensive change by the incumbent. Entry is *deterred* if the bank deliberately chooses an alternative (R_b, I, M_b) making the entrant's maximal operating profit nonpositive. Entry is *accommodated* when the bank chooses a commitment that leaves strictly profitable platform lending. With no fixed entry cost, and because no lending yields zero, a zero-profit entrant is indifferent between entering with zero mass or not lending; the no-entry outcome must be selected explicitly at a tie.

Let $\widehat{\Pi}_p(R_b, I, M_b; \mu) = \sup_{R_p, M_p \neq \varnothing} \Pi_p$ denote the platform's maximum operating profit under a nonempty lending policy. Define Π_b^D as the bank's best profit subject to deterring entry, and Π_b^A as its equilibrium profit under accommodation:

$$\Pi_b^D = \sup_{R_b, I, M_b : \widehat{\Pi}_p \leq 0} \Pi_b^M(R_b, I, M_b).$$

The bank obtains the corresponding bank-only allocation when the entrant chooses nonentry; at a zero-profit tie this requires specifying nonentry as the follower's selection. The bank has an incentive to deter only if $\Pi_b^D \geq \Pi_b^A$.

The decision admits a useful decomposition:

$$\Pi_b^D - \Pi_b^A = (\Pi_b^{M,*} - \Pi_b^A) - (\Pi_b^{M,*} - \Pi_b^D).$$

The first difference measures the recoverable monopoly advantage; the second is the loss in bank-only profit caused by the defensive commitment.

A deterrent rate or screening choice may destroy substantial bank-only profit, even if eliminating the platform restores the bank's exclusive lending opportunities.

At $\mu = 0.10$ under the benchmark $\alpha = 2$ and $c_b = 1$, the unadjusted monopoly optimum is $R_b^M \simeq 1.7522$, $I^M \simeq 0.7861$, and $\Pi_b^{M,*} \simeq 0.2080$. The platform then has strictly positive attainable profit, so entry is not blockaded. The accommodating equilibrium instead yields $\Pi_b^A \simeq 0.2038$. An illustrative aggressive deterrent commitment $R_b = 1.20$, $I = 0.80$ leaves the bank only 0.0077, but it is not the optimal deterrent policy.

A numerical constrained search over bank rates and screening intensity, retaining the exact piecewise platform best response and comparing all bank approval policies, locates a substantially better near-binding deterrent commitment at $R_b \simeq 1.3453$ and $I \simeq 0.8453$. Its bank-only profit is approximately $\Pi_b^D \simeq 0.0662$; the platform's maximum positive-policy return is approximately zero, and nonentry must be chosen at an exact tie. Even this improved deterrent payoff is far below accommodation. The recoverable monopoly advantage is only about 0.0042, against approximately 0.1418 in sacrificed bank-only profit at the deterrent commitment. Thus, in this numerical case, preserving the lending relationship is not worth the distorted rate and screening policy needed to exclude the platform. The search supports the local numerical comparison but is not a global analytical characterization of deterrence for every adoption rate or approval policy. In particular, feasibility at $\mu = 0.10$ does not imply feasibility at higher adoption, when the platform's signal becomes more informative.

B.8 Supporting implications: adoption observability and bank portfolio risk

If adoption were observable and the bank could post R_b^A and R_b^N to adopters and nonadopters, but retained common screening and favorable-signal lending, the bank's objective in the

benchmark cell would be

$$(1-\mu)G_M(R_b^N, I) + \mu\big[G_M(R_b^A, I) - a_S\delta\Delta_S^A m_b^S(R_b^A)\big] - C(I),$$

where $G_M(R, I) = \Pi_b^M(R, I) + C(I)$ is the bank's gross monopoly return. Holding I fixed, the nonadopter rate obeys $F_M(R_b^N, I) = 0$, while the adopter rate satisfies the competitive condition without the multiplier μ. This shows which common-price restriction is responsible for the spillover of competitive repricing to nonadopters. It does *not* imply that both rates, or a loan-weighted average, fall. The calibrated constrained exercise at $\mu = 0.50$ gives $R_b^N \simeq 1.752$, $R_b^A \simeq 2.038$ and bank portfolio default frequency about 10.18%, compared with the pooled bank rate 1.836 and default frequency 10.55%. This is a restricted pricing extension, not an equilibrium with unrestricted segment-specific screening and approval policies.

Finally, at unchanged bank choices in the benchmark coverage region, entry displaces $d = \mu a_S\delta\Delta_S$ safe bank loans. The bank's origination masses change from (L_0^S, L_0^R) to $(L_0^S - d, L_0^R)$, a special case of the general relative-displacement formula in Appendix B.10.

The benchmark equilibrium path is computed at 1,001 values of $\mu \in [0.10, 0.90]$ under identical primitives for monopoly and competition. All reported points satisfy the safe-covered/risky-uncovered inequalities. The follower's rate is checked against the exact piecewise pricing cells and all offer policies at each point; independent multistart bank comparisons are available at 21 adoption values. The agreement of analytical and finite-difference derivatives supports the smooth-branch FOCs, but a finite numerical search is not an analytical uniqueness proof. Delayed screening, simultaneous pricing, and screening-signal-contingent price discrimination are excluded from this appendix because they require distinct games and would not sharpen the maintained benchmark's equilibrium characterization.

B.9 Smooth undercutting and the informational allocation externality

At fixed approval policies within the interior two-offer region, a bank rate decrease of $\varepsilon > 0$ shifts the safe-borrower cutoff by $\theta^S \varepsilon/(2t)$ if the platform rate is fixed. Along a differentiable platform best response, the shift is $\theta^S(1-\rho_p)\varepsilon/(2t) + o(\varepsilon)$. Multiplying by $\mu p \gamma \delta$ gives the first-order mass of safe borrowers won from the platform. This mass converges to zero with ε, unlike the discrete transfer of all jointly approved applicants at a tied bid in Broecker (1990). This smoothness does not exclude follower-policy switches or multiple global best responses at non-interior points.

B.10 Relative displacement, portfolio quality, and data precision

Let $p_S = p$ and $p_R = 1 - p$, and retain the general type-conditional offer probabilities q_b^j, q_p^j introduced above. Define the bank-only mass $L_0^j = p_j q_b^j D_b^j$ at the same bank rate and approval policy as the competitive outcome. The ex ante loan-mass identities imply

$$L_b^j = L_0^j - \mu p_j q_b^j q_p^j \Delta_j = L_0^j(1 - s_j), \qquad s_j = \frac{L_0^j - L_b^j}{L_0^j} = \frac{\mu q_p^j \Delta_j}{D_b^j}, \quad \Delta_j = D_b^j - \widetilde{D}_b^j \geq 0.$$

The proportional displacement s_j is defined for $D_b^j > 0$. Because $0 \leq \widetilde{D}_b^j \leq D_b^j \leq 1$, the bank cannot originate more loans of either true type at unchanged bank decisions following entry. The statement does not rank separately optimized monopoly and competitive bank loan books.

Let $T = L_0^S + L_0^R$ and $Q = L_b^S + L_b^R$. The default-frequency identity $\Omega_b = (1 - \theta^S) + (\theta^S - \theta^R) L_b^R / Q$ gives

$$\begin{aligned} \Omega_b - \Omega_{b,0} &= (\theta^S - \theta^R)\left[\frac{L_0^R(1 - s_R)}{Q} - \frac{L_0^R}{T}\right] \\ &= (\theta^S - \theta^R)\frac{L_0^S L_0^R (s_S - s_R)}{TQ}. \end{aligned}$$

This establishes the first part of Proposition 1 for positive type-specific originations.

Under favorable-signal lending $q_p^S = \delta$, $q_p^R = 1 - \delta$, so at fixed (R_b, R_p, I),

$$\frac{\partial L_b^S}{\partial \delta} = -\mu p \gamma \Delta_S \leq 0, \qquad \frac{\partial L_b^R}{\partial \delta} = \mu(1-p)(1-\gamma)\Delta_R \geq 0.$$

As $\theta^S > \theta^R$, differentiation of the risky-origination share yields

$$\frac{\partial \Omega_b}{\partial \delta} = (\theta^S - \theta^R) \frac{L_b^S \, \partial_\delta L_b^R - L_b^R \, \partial_\delta L_b^S}{(L_b^S + L_b^R)^2} \geq 0.$$

The inequality is strict when both types are financed and either Δ_S or Δ_R is positive. This is a *fixed-decision* comparative static, not an assertion that equilibrium risk is monotone in platform precision. For the maintained technology $\delta(\mu; \alpha) = (1 + \mu^{1/\alpha})/2$ and $0 < \mu < 1$, one has $\delta_\alpha = -\mu^{1/\alpha} \log \mu / (2\alpha^2) > 0$. Thus the precision result also applies to an increase in α at fixed adoption and lending decisions.

In the benchmark, $\Delta_R = 0$ and $d = \mu p \gamma \delta \Delta_S$, so the general identity reduces to

$$\Omega_b - \Omega_{b,0} = \frac{(\theta^S - \theta^R) d L_0^R}{T(T-d)} \geq 0.$$

The direct diversion term has derivative $\partial_\mu(\mu\delta) = \delta + \mu\delta_\mu$. This separates the platform's reach effect from the additional effect of adoption-driven signal improvement. It is distinct from the total equilibrium derivative, which also includes changes in the two rates and screening.

Full-coverage robustness. The relative-displacement criterion does not require uncovered risky demand. To demonstrate this numerically without relaxing negative-NPV risky projects, set $Y^R = 0.90$ and retain the other benchmark primitives. The previously computed 801-point candidate equilibrium path for $\mu \in [0.10, 0.90]$ has strictly overlapping acceptable offers for both types; selective lending remains optimal on the globally checked points. Bank default frequency rises from approximately 10.38% in its own monopoly to

13.43% at $\mu = 0.90$, with safe loans displaced proportionately more than risky ones. The exact platform policy was checked at every grid point and independent bank global policy comparisons agreed at nine points. These are robustness calculations, not a theorem of global uniqueness.

B.11 Why the conditional competitive bank rate is higher

Consider the strictly interior safe-covered/risky-uncovered cell under favorable-signal lending. Let $a_S = p\gamma$ and $a_R = (1-p)(1-\gamma)$, and write $m_b^j = \theta^j R_b - c_b$. The derivative of monopoly profit with respect to R_b, at fixed I, is

$$F_M(R_b, I) = \sum_{j \in \{S,R\}} a_j \theta^j \left[D_b^j - \frac{m_b^j}{t} \right].$$

At the conditional monopoly optimum, $F_M(R_b^M(I), I) = 0$. If $a_R > 0$, $D_b^R > 0$, and $m_b^R < 0$, the risky term in this condition is strictly positive. Consequently the safe term is negative:

$$D_b^S(R_b^M) < \frac{m_b^S(R_b^M)}{t}.$$

Since $\bar{x}^S = (\bar{x}_b^S + \bar{x}_p^S)/2$, the displacement term under interior demands satisfies

$$\Delta_S = D_b^S - \widetilde{D}_b^S = \frac{D_b^S - \bar{x}_p^S}{2} < \frac{D_b^S}{2} < \frac{m_b^S}{2t},$$

where $\bar{x}_p^S > 0$ follows from the platform-only safe demand being strictly smaller than one. Under nonnegative conditional platform rate pass-through, $\rho_p \geq 0$, this yields

$$\Delta_S - \frac{1+\rho_p}{2t} m_b^S < 0.$$

Differentiating the bank's profit wedge along the platform's best response,

$$\frac{d\Pi_b}{dR_b} = F_M(R_b, I) - \mu a_S \delta \theta^S \left[\Delta_S - \frac{1+\rho_p}{2t} m_b^S\right],$$

shows that the bank's marginal competitive profit is positive at the monopoly conditional rate. Strict concavity of the induced bank pricing objective *within that cell*, and the assumption that both optima are in the cell, imply $R_b^C(I) > R_b^M(I)$. The inequalities do not extend automatically to binding participation thresholds or to changes in approval policy. This is the conditional result asserted by Proposition 2.

B.12 The direct and strategic returns to screening

For the bank's gross monopoly return, write $\Lambda(R_b) = \{pD_b^S m_b^S - (1-p)D_b^R m_b^R\}/2$. With a fixed platform rate, the loan-mass identity implies

$$\frac{\partial(\Pi_b + C)}{\partial I} = \Lambda(R_b) - \frac{\mu p \delta}{2} \Delta_S m_b^S.$$

Because $\gamma'(I) = 1/2$ and Δ_S is independent of I at fixed posted rates, this establishes the fixed-rate direct screening displacement effect discussed in the main text. When the bank anticipates the platform's response, Δ_S also changes through $R_p^{\text{br}}(R_b, I; \mu)$, so

$$C'(I) = \underbrace{\Lambda(R_b)}_{\text{monopoly marginal benefit}} - \underbrace{\frac{\mu p \delta}{2} \Delta_S m_b^S}_{\text{direct displacement}} + \underbrace{\frac{\mu p \gamma \delta \theta^S}{2t} m_b^S \frac{\partial R_p^{\text{br}}}{\partial I}}_{\text{strategic repricing}}.$$

Define the entire right side as $\mathcal{H}(R_b, I, \mu)$. Differentiating the equilibrium screening condition yields

$$[C''(I^*) - \mathcal{H}_I] I_\mu^* = \mathcal{H}_\mu + \mathcal{H}_{R_b}(R_b^*)_\mu,$$

where the partial derivatives of $\mathcal{H}$ include any dependence of the platform's best response on μ and I. The denominator is positive at a strict conditional screening maximum. The turning

point occurs where the numerator vanishes; neither the sign of screening's first derivative nor the sign of its second derivative is a primitive implication of the data technology.

The benchmark equilibrium evaluation of the three right-hand-side terms confirms the interpretation in the main text: as μ approaches the high-adoption range, the safe-demand overlap shrinks enough that the direct displacement term becomes smaller in absolute value, while the strategic repricing contribution grows. The resulting screening path has an interior minimum. These observations are numerical properties of the maintained benchmark, not an analytic assertion about all parameter values.

C Adoption sorting and the distribution of lender convenience

The benchmark assumes a uniform unconditional convenience characteristic $X \sim U[0,1]$ and independent platform adoption. A more general environment allows the adoption decision to be correlated with lender convenience while preserving the distribution of borrower types and the platform's aggregate adoption rate. Let $U \sim U[0,1]$ index a latent adoption propensity, with $A = \mathbf{1}\{U \leq \mu\}$. Assume the borrower credit type $j \in \{S, R\}$ is independent of (X, U), and define their joint CDF (a copula, since both marginal distributions are uniform) as

$$C(x,u) = \Pr(X \leq x, U \leq u), \qquad (x,u) \in [0,1]^2. \tag{C.1}$$

We require that C have no point masses and be continuously differentiable at the interior cutoffs used below, with a nonnegative joint density. The benchmark is the independent copula $C_0(x,u) = xu$. At every $0 < \mu < 1$, the adopter and non-adopter conditional CDFs are

$$F_A(x;\mu) = \frac{C(x,\mu)}{\mu}, \qquad F_N(x;\mu) = \frac{x - C(x,\mu)}{1-\mu}, \quad \mu F_A + (1-\mu)F_N = x. \tag{C.2}$$

Thus the unconditional distribution of X stays uniform as the platform grows, and adoption is nested: users with $U \leq \mu$ remain adopters when μ increases. If adopters are weakly more platform-oriented than non-adopters in the sense of first-order stochastic dominance, then $F_A(x;\mu) \leq F_N(x;\mu)$, equivalently $C(x,\mu) \leq \mu x$. The independence case satisfies equality. Neither this ordering nor the marginal uniformity assumptions require a particular parametric CDF.

For a conventional smooth example within this class, let

$$C_\rho(x,u) = \Phi_2\big(\Phi^{-1}(x), \Phi^{-1}(u); \rho\big), \qquad -1 < \rho < 1,$$

where Φ_2 is the standard bivariate normal CDF with correlation ρ. For $\rho = 0$, $C_0(x,u) = xu$ and both conditional distributions are uniform. A negative ρ associates low adoption rank U with high platform convenience X, thereby sorting adopters toward the platform. This Gaussian-copula family serves only to demonstrate that the admissible class contains economically meaningful, non-independent distributions; no new equilibrium results rely on a chosen ρ.

C.1 Origination masses under correlated adoption

Conditional demands remain determined by borrower utility. With raw bank and platform acceptance cutoffs $\bar{x}_b^j, \bar{x}_p^j$ and the lender-indifference cutoff $\bar{x}^j$ as defined in the model, define the interval of borrowers diverted from the bank when both lenders extend credit:

$$\ell_j = \max\{0, \bar{x}_p^j, \bar{x}^j\}, \qquad h_j = \min\{1, \bar{x}_b^j\},$$

$$G_j(\mu) = \begin{cases} C(h_j,\mu) - C(\ell_j,\mu), & h_j > \ell_j, \\ 0, & h_j \leq \ell_j. \end{cases} \tag{C.3}$$

In the strictly interior covered region this reduces to $G_j = C(\bar{x}_b^j, \mu) - C(\bar{x}^j, \mu)$. Under independence, $G_j = \mu\Delta_j$, with $\Delta_j = D_b^j - \widetilde{D}_b^j$; for an uncovered type the interval is empty and $G_j = 0$.

Let $p_S = p$, $p_R = 1 - p$, and retain the type-conditional approval probabilities q_b^j, q_p^j defined in Appendix B. Because true type is independent of (X, U), and because the bank may serve all borrowers, its bank-only origination mass at unchanged decisions remains $L_0^j = p_j q_b^j D_b^j$. Its competitive mass is exactly

$$L_b^j = p_j q_b^j \big[D_b^j - q_p^j G_j(\mu)\big]. \tag{C.4}$$

To obtain the platform's mass, define $v_j = [\bar{x}_p^j]_0^1$ and the interval of adopters who accept the platform alone but prefer the bank when two offers are available:

$$a_j = \max\{0, \bar{x}_p^j\}, \quad b_j = \min\{1, \bar{x}_b^j, \bar{x}^j\}, \quad H_j(\mu) = \begin{cases} C(b_j, \mu) - C(a_j, \mu), & b_j > a_j, \\ 0, & b_j \le a_j. \end{cases}$$

Then

$$L_p^j = p_j q_p^j \big[\mu - C(v_j, \mu) - q_b^j H_j(\mu)\big]. \tag{C.5}$$

These equations integrate the joint probabilities of true type, adoption, approval and lender choice. Ex ante profits remain $\Pi_b = \sum_j L_b^j(\theta^j R_b - c_b) - C(I)$ and $\Pi_p = \sum_j L_p^j(\theta^j R_p - c_p)$; no posterior is multiplied into an already type-specific loan mass. Bank application demand, $p_j D_b^j$, is unchanged because the *unconditional* distribution of X is still uniform. Platform application demand is instead $p_j[\mu - C(v_j, \mu)]$, so its applicant priors and optimal pricing generally change.

Lemma 2 (Relative displacement with sorted adoption)**.** *At unchanged posted rates, screening and approval policies, suppose $D_b^S, D_b^R > 0$ and the bank originates positive loans of both types. Let $s_j \equiv q_p^j G_j(\mu)/D_b^j$. Then $L_b^j = L_0^j(1 - s_j)$ and the bank's expected default frequency*

increases relative to bank-only lending if and only if $s_S > s_R$. In a favorable-signal equilibrium, an increase in platform accuracy δ holding all other decisions fixed weakly reduces L_b^S and weakly increases L_b^R, and therefore weakly increases the bank's expected default frequency. The statements include independent adoption as the special case $C(x,u) = xu$.

Proof. Equation (C.4) immediately gives $L_b^j = L_0^j(1-s_j)$. Let $T = L_0^S + L_0^R$ and $Q = L_b^S + L_b^R$. The definition $\Omega_b = [(1-\theta^S)L_b^S + (1-\theta^R)L_b^R]/Q$ gives

$$\Omega_b - \Omega_{b,0} = (\theta^S - \theta^R)\frac{L_0^S L_0^R (s_S - s_R)}{TQ},$$

which proves the sign claim. For favorable-signal approval, $q_p^S = \delta$ and $q_p^R = 1-\delta$, so $\partial_\delta L_b^S = -p\gamma G_S \leq 0$ and $\partial_\delta L_b^R = (1-p)(1-\gamma)G_R \geq 0$. More explicitly, $\partial_\delta \Omega_b = (\theta^S - \theta^R)[L_b^S \partial_\delta L_b^R - L_b^R \partial_\delta L_b^S]/(L_b^S + L_b^R)^2 \geq 0$, proving the remaining claim. □

C.2 Reach, information, and marginal price competition

In the interior safe-covered region the bank's safe-loan diversion at fixed lender decisions is

$$d_S = p\gamma\delta(\mu;\alpha)G_S(\mu), \qquad G_S(\mu) = C(\bar{x}_b^S, \mu) - C(\bar{x}^S, \mu).$$

Along a fixed adoption-ranking copula, and holding the cutoffs fixed,

$$\frac{\partial d_S}{\partial \mu} = p\gamma \left[\underbrace{\delta_\mu G_S}_{\text{information improvement}} + \underbrace{\delta\{C_u(\bar{x}_b^S, \mu) - C_u(\bar{x}^S, \mu)\}}_{\text{new adopters in the contestable interval}} \right] \geq 0. \tag{C.6}$$

The second term is nonnegative since $C_u(h,\mu) - C_u(\ell,\mu)$ is the conditional probability that a marginal adopter lies between the two cutoffs. Under independence, $G_S = \mu\Delta_S$ and the expression reduces to $p\gamma\Delta_S(\delta + \mu\delta_\mu)$. In particular, a larger platform need not displace the same share of safe bank loans as an equally large platform with uniformly distributed users.

If F_A has a continuous density $f_A(x;\mu) = C_x(x,\mu)/\mu$, an infinitesimal bank-rate cut $\varepsilon > 0$

along a differentiable platform best response wins a mass of jointly approved safe adopters equal to

$$\mu p\gamma\delta\, f_A(\bar{x}^S;\mu)\frac{\theta^S(1-\rho_p)}{2t}\varepsilon + o(\varepsilon),$$

provided the indifference cutoff stays in the strictly covered interior. Local price competition depends on the *density of adopters at the contestable margin*, not merely their total mass. First-order stochastic dominance of adopters' X over nonadopters' X does not generally order G_S or $f_A(\bar{x}^S)$ relative to the independence benchmark: interval probabilities and point densities depend on the shape of the CDF, not only its first-order ranking.

Although the absolute continuity assumption prevents an infinitesimal cut from transferring a positive mass of borrowers at once, it does not ensure that the platform's profit remains quadratic or that its best response remains unique. With sorted adoption both lenders' equilibrium problems must therefore be re-solved using (C.4)–(C.5); the benchmark's affine follower best response and interior existence arguments cannot be reused mechanically. This extension establishes the robustness of the *conditional* displacement logic, not a newly computed equilibrium path under correlated adoption.

D Platform information and funding-cost comparative statics

D.1 Proof of Proposition 3

For favorable-signal lending, let $\Delta_j = D_b^j - \widetilde{D}_b^j \geq 0$ and $m_b^j = \theta^j R_b - c_b$. The ex ante loan masses in (5.1) can be written without additional posterior probabilities as

$$L_b^S = L_0^S - \mu p\gamma\delta\Delta_S, \qquad L_b^R = L_0^R - \mu(1-p)(1-\gamma)(1-\delta)\Delta_R,$$

where L_0^j is evaluated at the same bank rate and screening as under competition. Holding R_b, R_p, I fixed, $\delta_\alpha = -\mu^{1/\alpha}\log\mu/(2\alpha^2) > 0$ and

$$\partial_\alpha L_b^S = -\mu p\gamma\Delta_S\delta_\alpha, \qquad \partial_\alpha L_b^R = \mu(1-p)(1-\gamma)\Delta_R\delta_\alpha.$$

With $\Omega_b = (1-\theta^S) + (\theta^S - \theta^R)L_b^R/(L_b^S + L_b^R)$, differentiation gives

$$\left.\frac{\partial\Omega_b}{\partial\alpha}\right|_{R_b,R_p,I} = \frac{(\theta^S-\theta^R)\mu\delta_\alpha}{(L_b^S+L_b^R)^2}\left[p\gamma\Delta_S L_b^R + (1-p)(1-\gamma)\Delta_R L_b^S\right] \geq 0. \tag{D.1}$$

The inequality is strict when both types are originated and at least one of Δ_S, Δ_R is positive. Consequently the result does not rely on an uncovered risky market. The bank's profit derivative is

$$\left.\frac{\partial\Pi_b}{\partial\alpha}\right|_{R_b,R_p,I} = -\mu\delta_\alpha\left[p\gamma\Delta_S m_b^S - (1-p)(1-\gamma)\Delta_R m_b^R\right] \leq 0$$

when $m_b^S \geq 0$ and $m_b^R \leq 0$. Since $\gamma'(I) = 1/2$ and the rate-dependent quantities are held fixed, its gross marginal screening return satisfies

$$\left.\frac{\partial}{\partial\alpha}\left(\frac{\partial\Pi_b}{\partial I} + C'(I)\right)\right|_{R_b,R_p} = -\frac{\mu\delta_\alpha}{2}\left[p\Delta_S m_b^S - (1-p)\Delta_R m_b^R\right] \leq 0. \tag{D.2}$$

In the benchmark's risky-uncovered case, $\Delta_R = 0$, reducing (D.1) to (6.1). The proof does not assume that the derivatives retain these signs after lenders reoptimize. □

D.2 Interior pass-through and the safe-coverage frontier

Lemma 3 (Conditional funding-cost pass-through)**.** *Within the strictly interior safe-covered/risky-uncovered region, the bank's conditional optimal rate increases with c_b when $c_p = \psi c_b$, for fixed screening, adoption and information quality.*

Proof. The platform's interior response is affine, $R_p = a_0 + \eta_p c_p + \rho_p R_b$, with $\eta_p > 0$ and $0 < \rho_p < 1$. At fixed I, μ, α, bank loan masses can then be written $L_b^j = A_j - K_j R_b$, $K_j > 0$. The

risky-loan intercept does not vary with funding costs, while $\partial A_S/\partial c_b = \psi p\gamma\mu\delta\theta^S\eta_p/(2t) > 0$. The bank rate solves

$$R_b^C(I) = \frac{\theta^S A_S + \theta^R A_R + c_b(K_S + K_R)}{2(\theta^S K_S + \theta^R K_R)}.$$

The slope coefficients K_j are independent of funding costs in this cell, so

$$\left.\frac{\partial R_b^C}{\partial c_b}\right|_I = \frac{K_S + K_R + \psi p\gamma\mu\delta(\theta^S)^2\eta_p/(2t)}{2(\theta^S K_S + \theta^R K_R)} > 0.$$

The argument does not apply after a participation or approval-policy boundary is reached.

For interior participation thresholds, the share of safe bank-only demand lost when both offers are acceptable is

$$\Delta_S = \left[\frac{2Y^S - \theta^S(R_b + R_p) - t}{2t}\right]_+,$$

within the relevant pricing cell. The bracketed formula is not meant to replace the general truncated-demand definition outside this cell. The coverage frontier is $\theta^S(R_b+R_p) = 2Y^S - t$. At the frontier, profit is continuous but the marginal return to a rate change is determined by its one-sided derivatives. The exact follower optimum and the bank's other lending policies must still be compared.

The supplied numerical exercise traces 2,001 adoption values in $[0.10, 0.90]$ for each of eight combinations of $\alpha \in \{1.25, 1.50, 1.75, 2\}$ and $c_b \in \{1, 1.1\}$, holding $c_p = 1.1c_b$. The platform's piecewise best response and approval policy are checked at all 16,008 points; independently optimized bank-policy checks are available at selected locations. At $c_b = 1$, all traced paths remain strictly interior. At $c_b = 1.1$, the safe-coverage frontier first binds at $\mu \simeq 0.7952, 0.7916, 0.7884, 0.7864$, respectively. These are verified numerical equilibrium candidates at grid points, not an analytical uniqueness theorem for every adoption value.

D.3 Additional figures and numerical comparisons

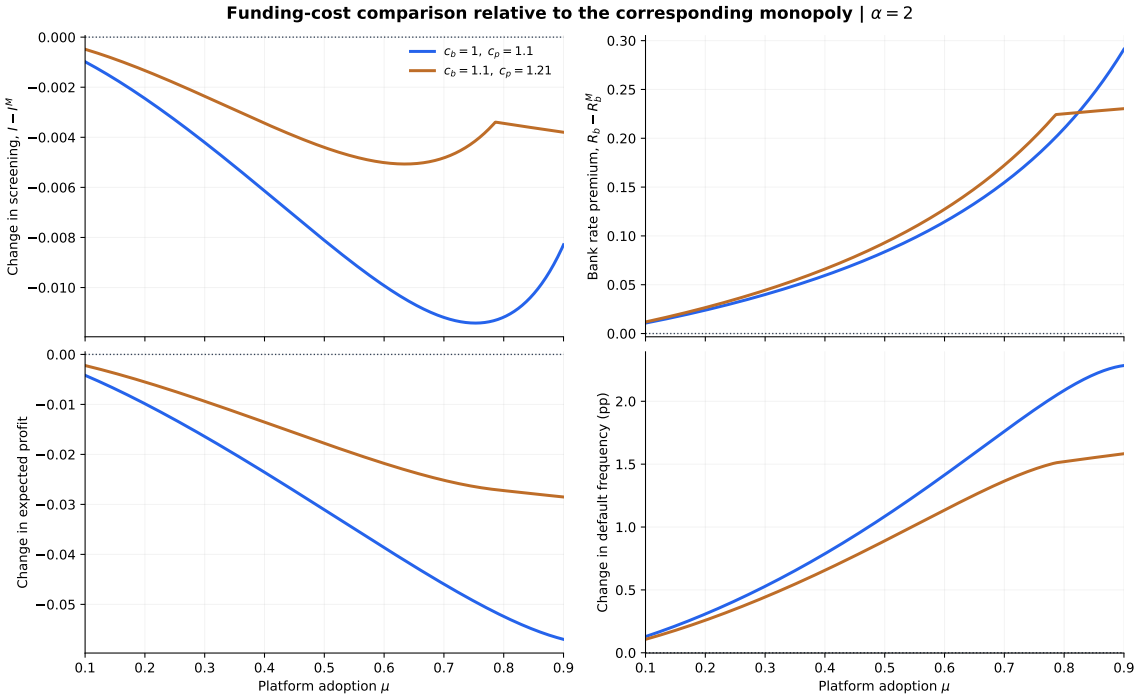


Figure 11: **Funding-cost responses relative to monopoly** ($\alpha = 2$). Differences are computed relative to the reoptimized monopoly under each respective funding cost, avoiding comparisons with different primitives.

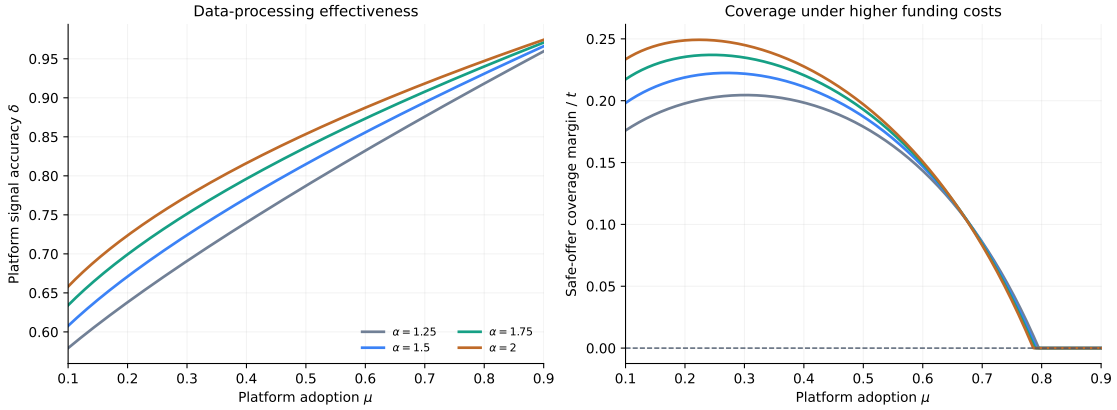


Figure 12: **Information precision and acceptable-offer overlap.** Platform information quality and the safe-market coverage frontier across adoption levels and funding-cost configurations.

E Efficiency and welfare accounting

Productive efficiency. The social return of a type-j loan made by intermediary ℓ is $Y^j - c_\ell$. The model assumes $Y^S > c_p > c_b > Y^R$ and no aggregate bank lending-capacity limit. A planner who observes true borrower types therefore lends only to safe types, uses the bank, and incurs no screening cost. Thus $W^{P,FB} = p(Y^S - c_b)$. Subtracting equilibrium productive surplus in (7.1) and collecting terms by type yields the decomposition in (7.2). In particular, the same expected project payoff Y^j appears in the private loan margins and the social surplus; failure is already reflected in expected output.

Borrower convenience and total welfare. For the favorable-signal policies, let $p_S = p, p_R = 1 - p$ and denote the probability of approval conditional on true type by q_b^j, q_p^j, as in Appendix B. Under the uniform convenience distribution, the ex ante convenience cost incurred by borrowers who obtain bank-only, platform-only, and competing offers is

$$\begin{aligned}\mathcal{D} = \frac{t}{2} \sum_{j\in\{S,R\}} p_j \Big\{ & q_b^j(1 - \mu q_p^j)(D_b^j)^2 + \mu q_p^j(1 - q_b^j)(D_p^j)^2 \\ & + \mu q_b^j q_p^j \big[(\widetilde{D}_b^j)^2 + (\widetilde{D}_p^j)^2\big] \Big\}.\end{aligned} \tag{E.1}$$

For a bank-only offer, for example, a financed borrower has $x_i \in [0, D_b^j]$, yielding $t\int_0^{D_b^j} x\,dx = t(D_b^j)^2/2$. Platform-only offers have the symmetric interval at the upper end of $[0, 1]$. When both offers are available, the bank and platform financing intervals do not overlap; their contributions are the squares of the two conditional demand lengths. The formula applies with the truncated demands used in the main model.

Borrower surplus equals

$$CS = \sum_{j\in\{S,R\}} \big[L_b^j(Y^j - \theta^j R_b) + L_p^j(Y^j - \theta^j R_p)\big] - \mathcal{D}.$$

Adding expected lender profits gives

$$CS + \Pi_b + \Pi_p = \sum_j \left[L_b^j (Y^j - c_b) + L_p^j (Y^j - c_p) \right] - C(I) - \mathcal{D} = W. \tag{E.2}$$

Expected repayments cancel; neither the screening expenditure nor the financing losses associated with default are counted twice. This identity presumes funding costs represent real resources, not purely financial transfers.

Efficient lender assignment. When convenience costs are real, a safe platform adopter with both lenders available should be assigned to the bank when $Y^S - c_b - tx \geq Y^S - c_p - t(1-x)$, i.e. $x \leq x^{\text{eff}} = \frac{1}{2} + (c_p - c_b)/(2t)$. At the benchmark parameters all safe projects are socially productive regardless of convenience, and the full-information total-surplus benchmark is

$$W^{FB}(\mu) = p \left(Y^S - c_b - \frac{t}{2} \right) + \mu p t (1 - x^{\text{eff}})^2.$$

For $p = 1/2$, $Y^S = 2.3$, $c_b = 1$, $c_p = 1.1$, and $t = 0.75$, this is $W^{FB}(\mu) \simeq 0.4625 + 0.07042\mu$. The increasing first-best is due solely to the possibility of assigning safe adopters with a high x to a more convenient lending channel. This result assumes that introducing a platform option does not itself consume resources.

Creation, diversion, and the incumbent response. At competitive bank decisions, hypothetical bank-only origination is $\widehat{L}_b^j = p_j q_b^j D_b^j$. Define $T^j = \widehat{L}_b^j - L_b^j \geq 0$ as loans diverted from the bank and $N^j = L_p^j - T^j \geq 0$ as newly funded projects, all evaluated at those same bank decisions. The difference in productive surplus relative to the *reoptimized* bank-only

equilibrium is then exactly

$$W^{P,C} - W^{P,M} = \sum_j (\widehat{L}_b^j - L_0^{j,M})(Y^j - c_b) + \sum_j N^j (Y^j - c_p) - \sum_j T^j (c_p - c_b) + C(I^M) - C(I^C).$$

The first term captures how the bank's own choices change its hypothetical lending portfolio; the second and third distinguish new platform credit from diverted bank lending. The identity follows by substituting $L_b^j = \widehat{L}_b^j - T^j$ and $L_p^j = N^j + T^j$. Since the hypothetical bank-only masses are evaluated at competitive decisions, this accounting identity is not a second equilibrium in which the bank reoptimizes after platform removal.

Numerical verification. The four components of (7.2) add to the calculated first-best gap at each of the 2,001 stored adoption values. The maximum absolute reconciliation error is below 3×10^{-16}. Net productive surplus reaches a minimum at $\mu = 0.1692$ and first exceeds the reoptimized bank-only benchmark at $\mu = 0.3472$ on the saved grid. Total welfare $W = W^P - \mathcal{D}$ increases along that same range. These are properties of the reported equilibrium branch on $[0.1, 0.9]$, not global comparative-static statements over all adoption values or market-coverage regimes.